

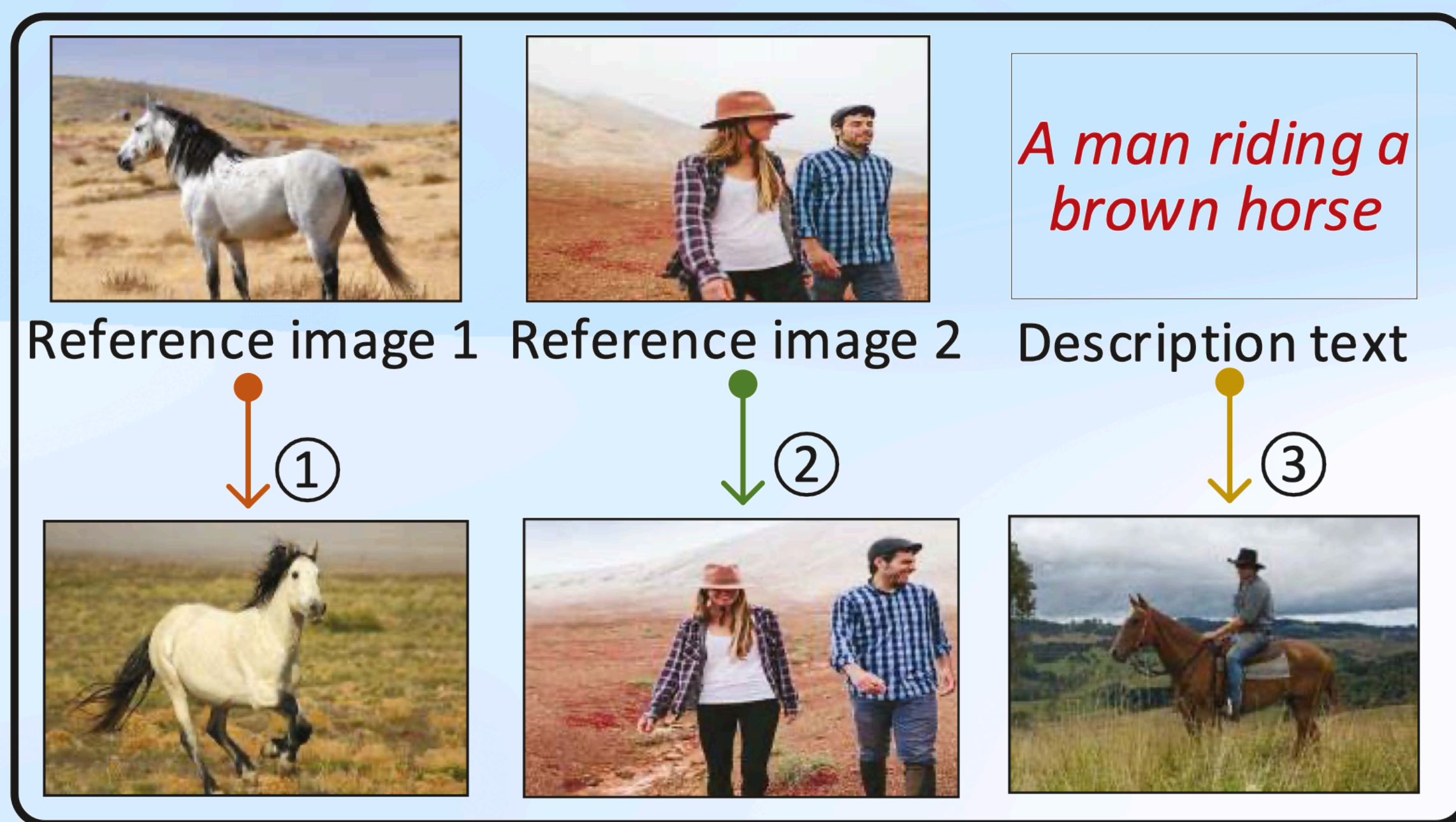
MUST - An Effective and Scalable Framework for Multimodal Search of Target Modality

Mengzhao Wang, Yunjun Gao, et al.
Zhejiang University, Hangzhou, China

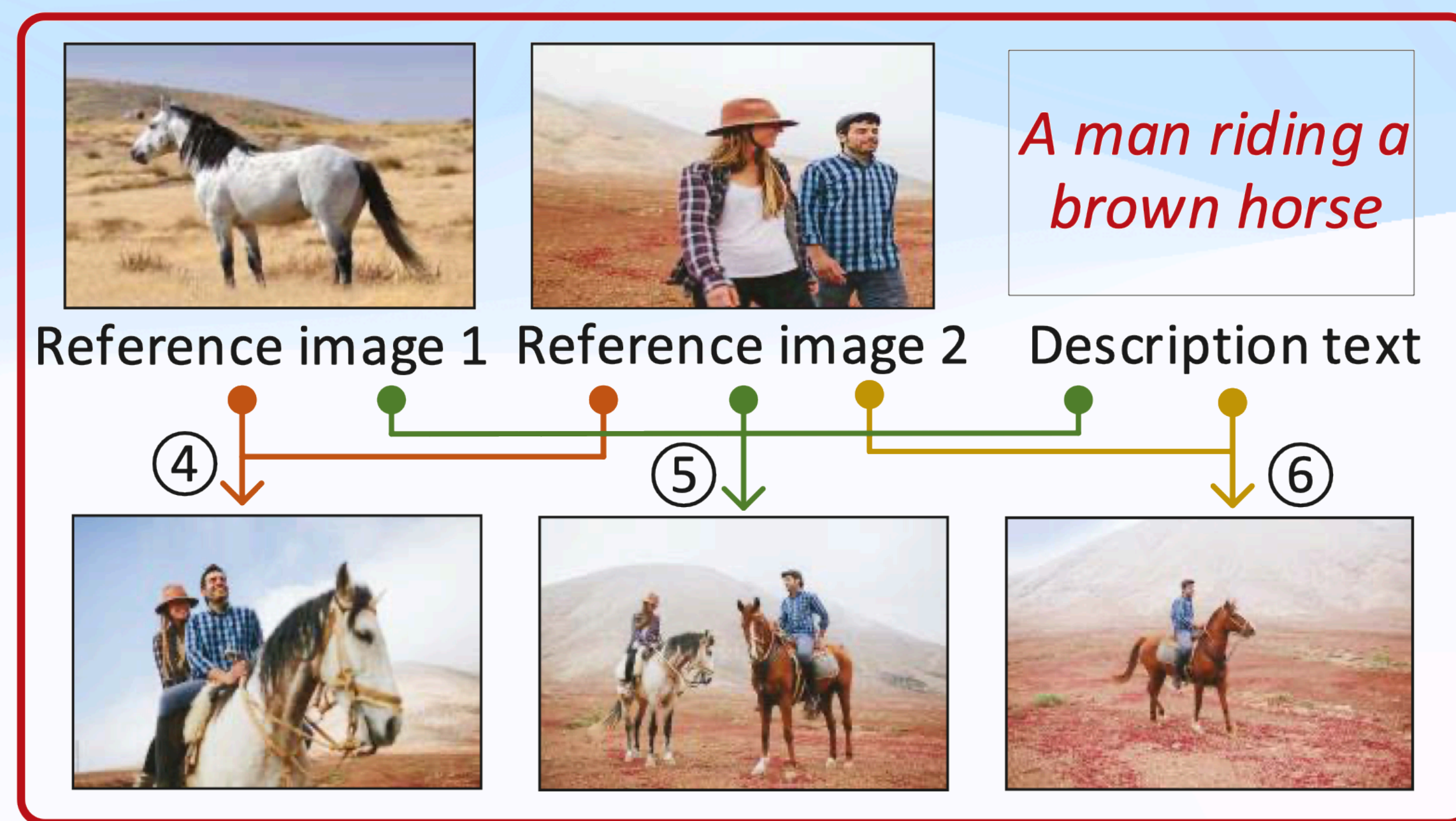
Multimodal Search of Target Modality (MSTM)

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Problem Illustration



(a) Traditional search paradigm



(b) Multimodal search paradigm

Multimodal Search of Target Modality (MSTM)

Problem Illustration / Formalization

Object Set $\mathcal{S}$. each $o \in \mathcal{S}$ has m modalities $o^i (0 \leq i \leq m - 1)$.

Query. A query q consists of $t \leq m$ modalities $q^i (0 \leq i \leq t - 1)$. q^0 represents the **target modality**.

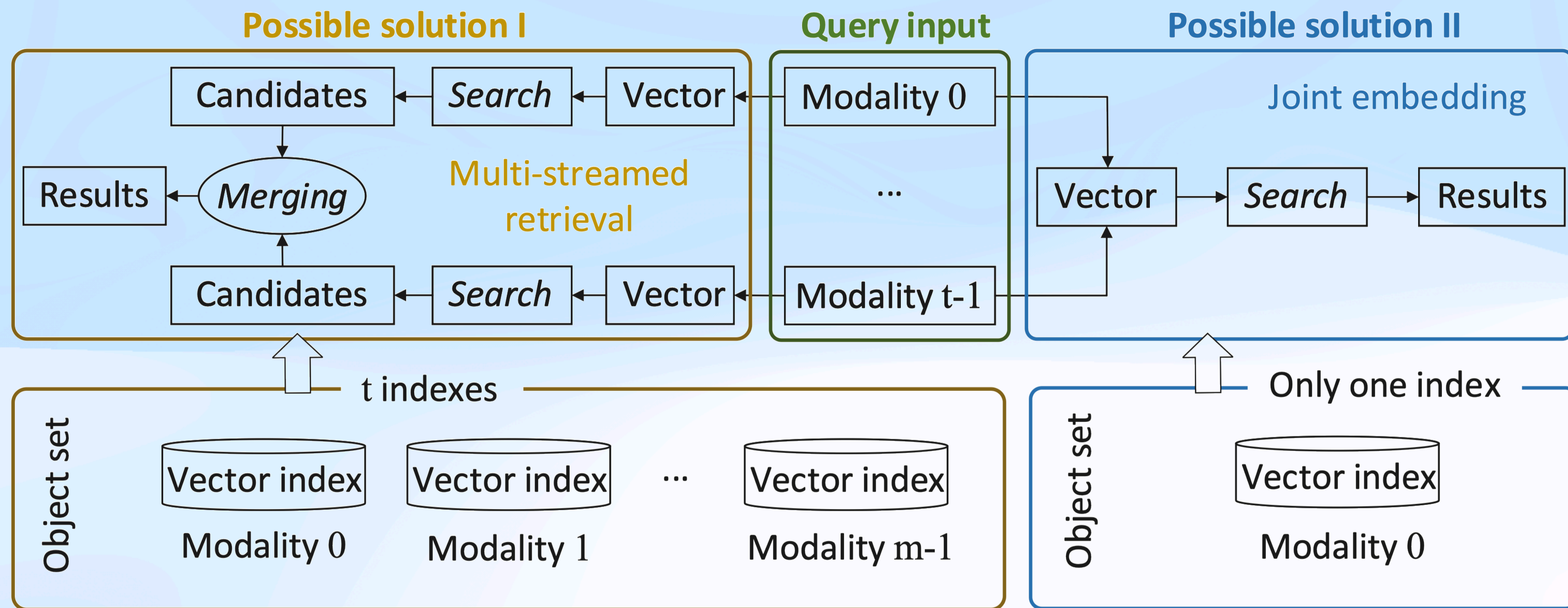
Problem. Given an object set $\mathcal{S}$, a query input q , and a positive integer k , the goal of **Multimodal Search of the Target Modality** problem is to find k objects $\mathcal{R}$ from $\mathcal{S}$ that best match the query.

Performance Metric (Recall Rate). Suppose there are k' ground-truth objects of q (denoted by $\mathcal{G}$) in the object set $\mathcal{S}$. The recall rate at k is formally defined as

$$\text{Recall@}k(k') = \frac{|\mathcal{R} \cap \mathcal{G}|}{k'}$$

Multimodal Search of Target Modality (MSTM)

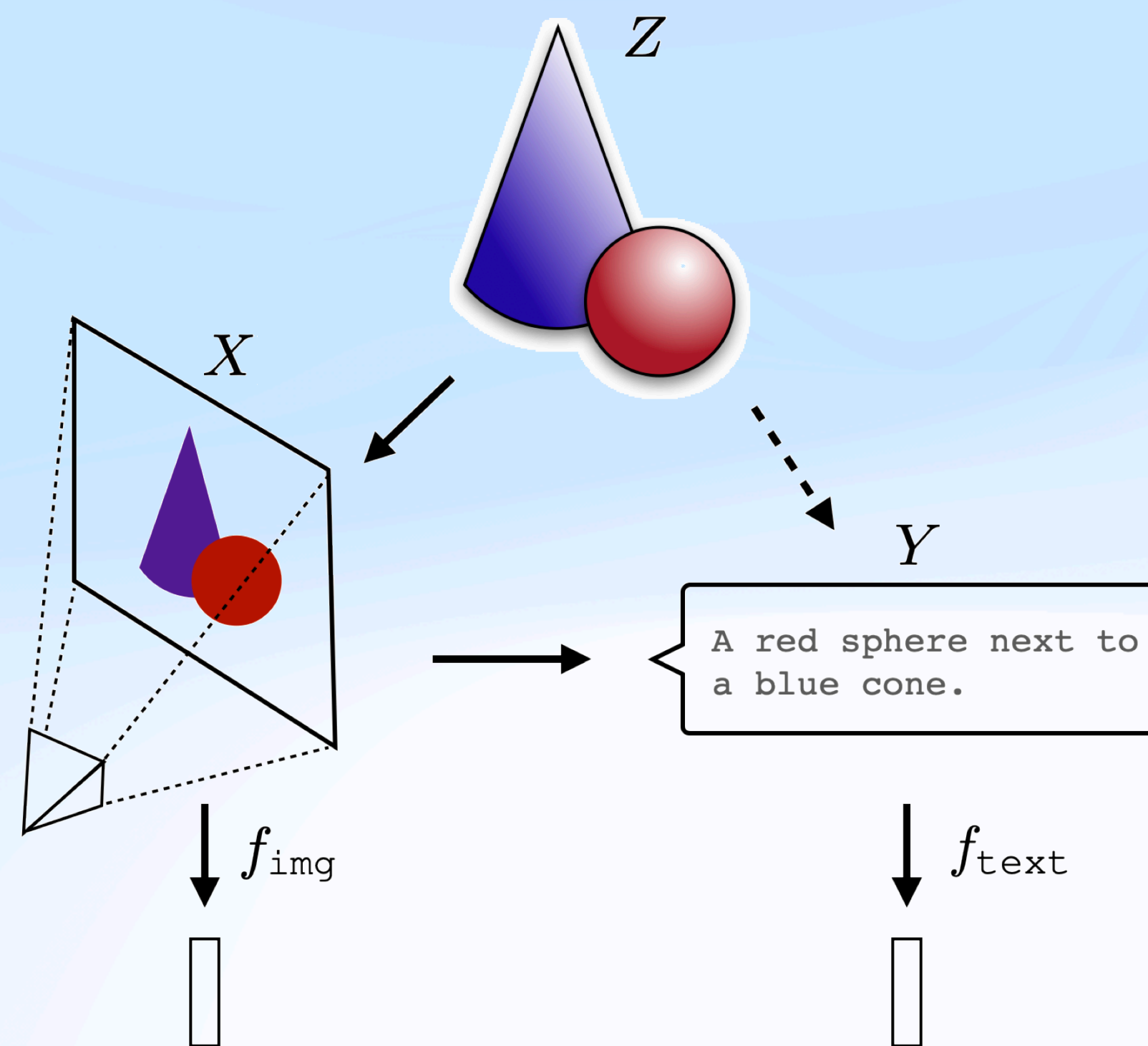
Baseline Solution



- **Multi-Stream Retrieval (MR).** Merge top candidates to get top results. (DB+IR)
- **Joint Embedding (JE).** Map multi-modal data into the same vector space. (CV)

Multimodal Search of Target Modality (MSTM)

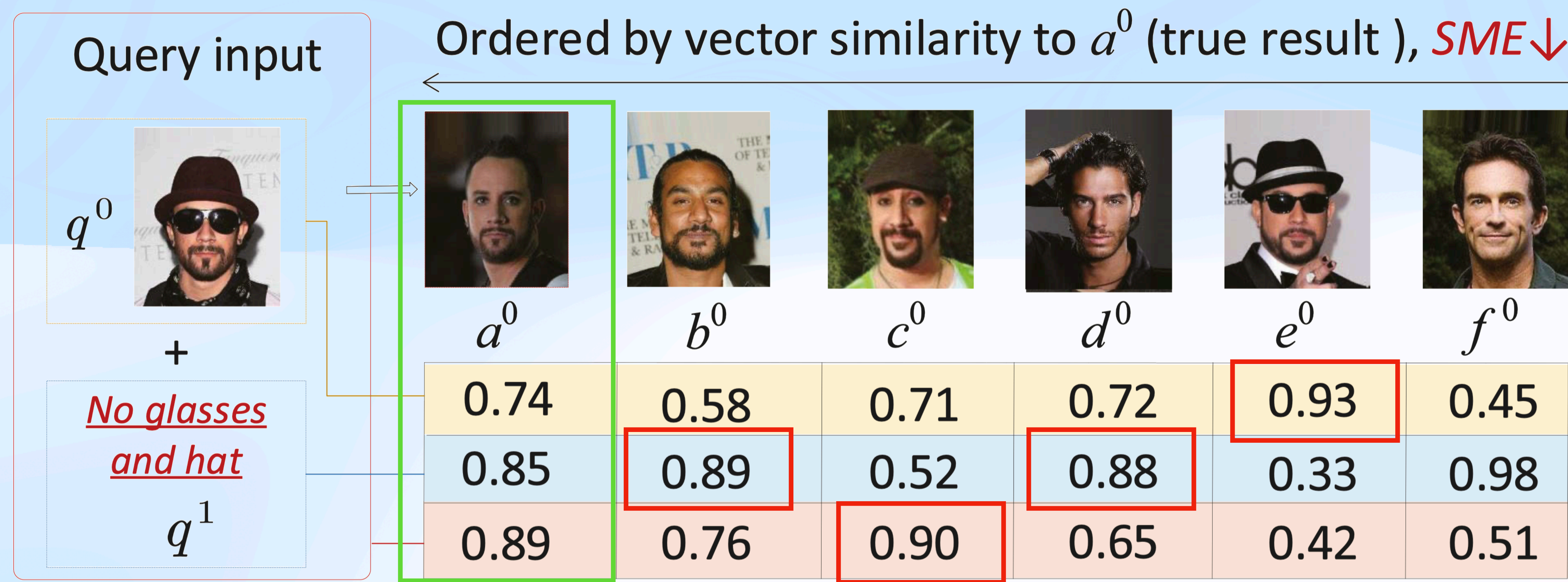
Baseline Solution / The Platonic Representation Hypothesis



Neural networks, trained with different objectives on different data and modalities, are converging to a **shared statistical model of reality** in their representation spaces.

Multimodal Search of Target Modality (MSTM)

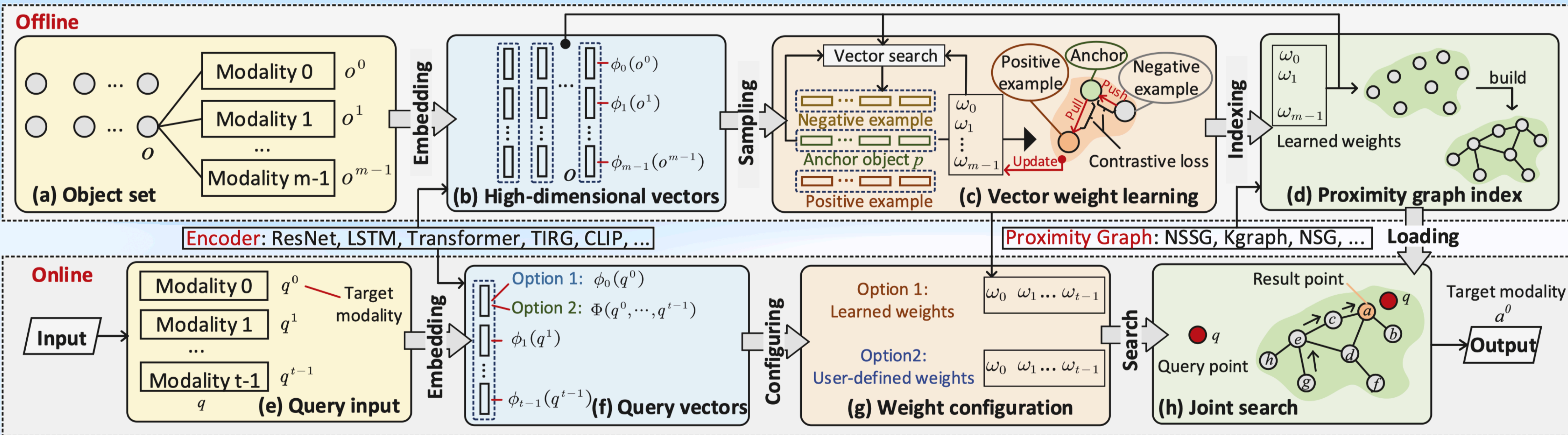
Baseline Solution / Why They Failed



MUST Framework

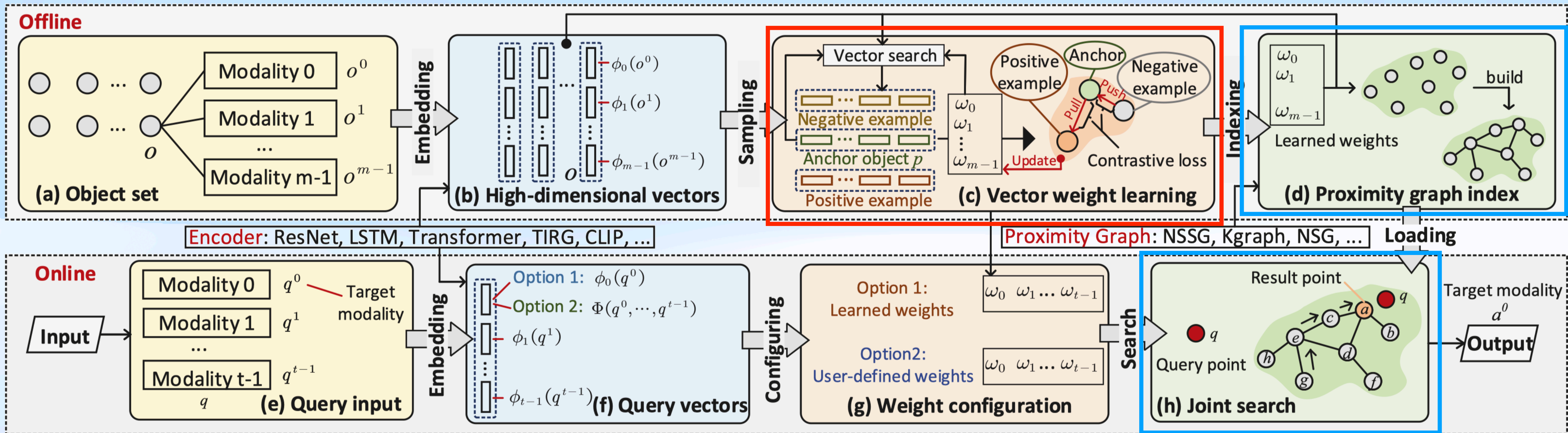
MUST Framework

Framework



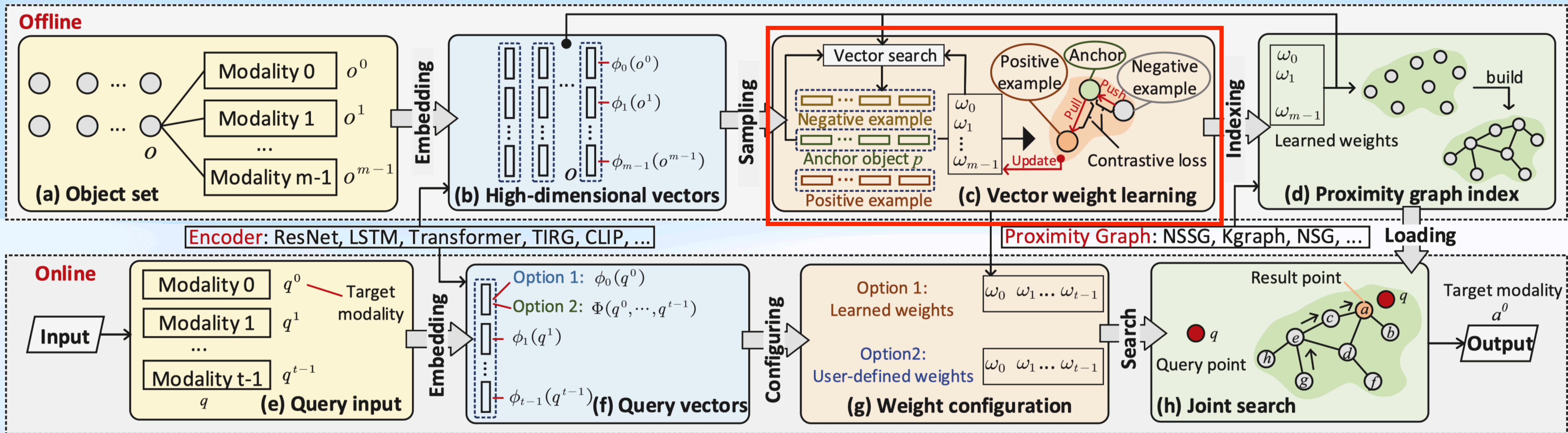
MUST Framework

Framework



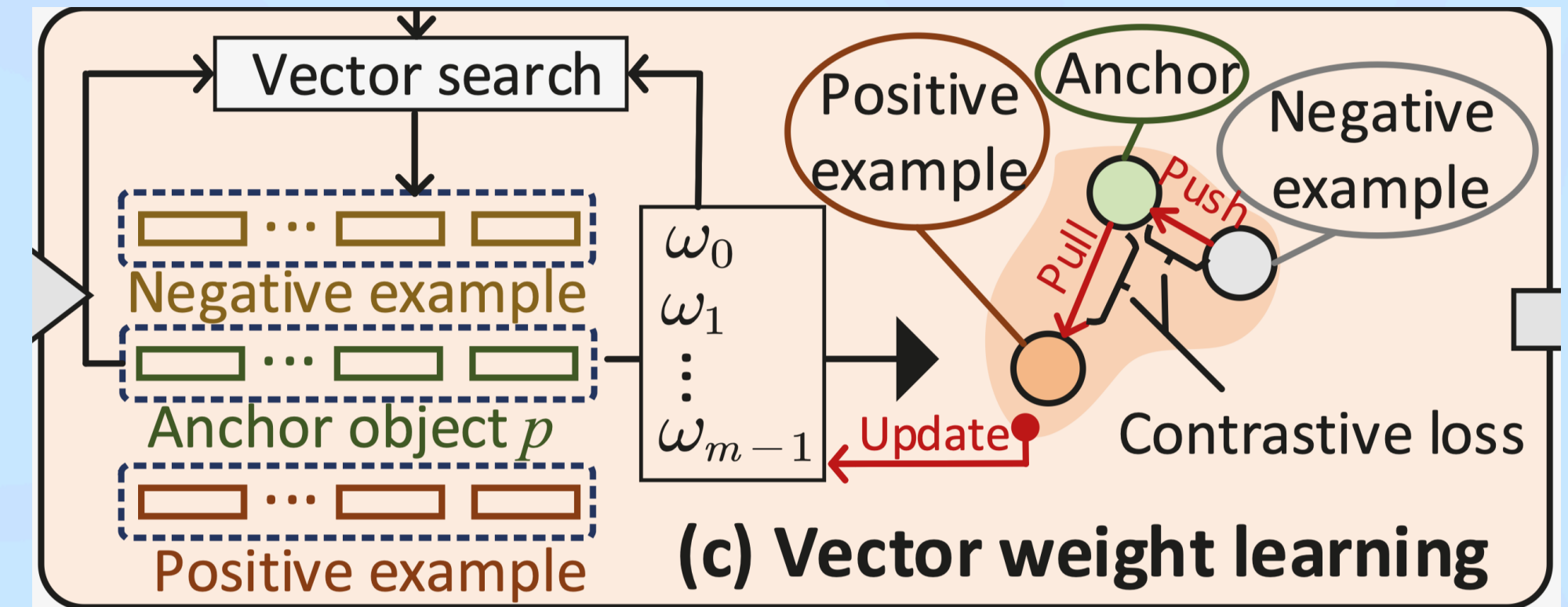
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Framework / Vector Weight Learning



MUST Framework

Vector Weight Learning



Concatenated vector for both data and query

$$\hat{p} = [\omega_0 \phi_0(p^0), \dots, \omega_{m-1} \phi_{m-1}(p^{m-1})]$$

- p^i is the data/query of i -th modality.
- ϕ_i is the encoder (embedding function) of i -th modality.
- w_i is the **learnable** weights of i -th modality.

MUST Framework

Vector Weight Learning

- **Positive Example.** Ground Truth.

- **Negative Example.** Vectors that are easily confused with the ground truth, that is, the top-K similarities (exclude the ground truth, if it does belongs to TopK). Formally:

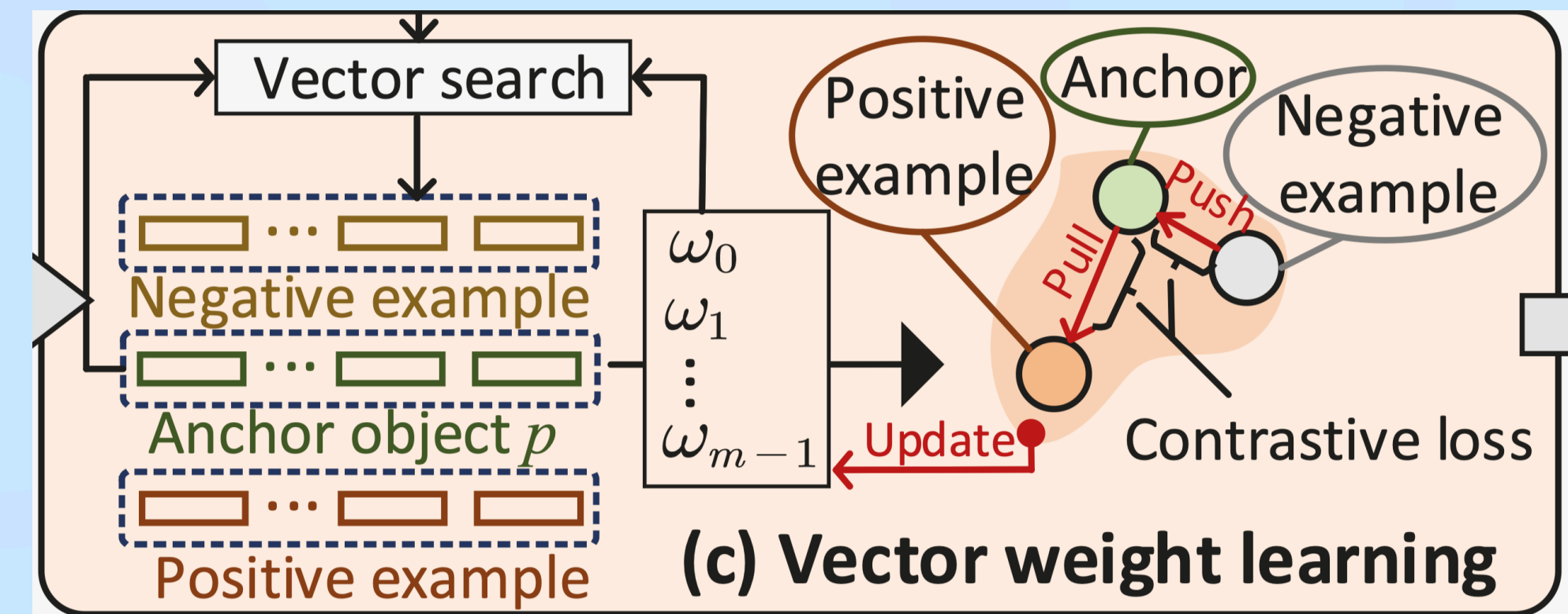
$$R = \arg \max_{R \subset T \wedge |R|=k} \sum_{r \in R} \hat{r} \hat{p}$$

$$N^+ = \{p^+\}$$

$$N^- = R \setminus \{p^+\}$$

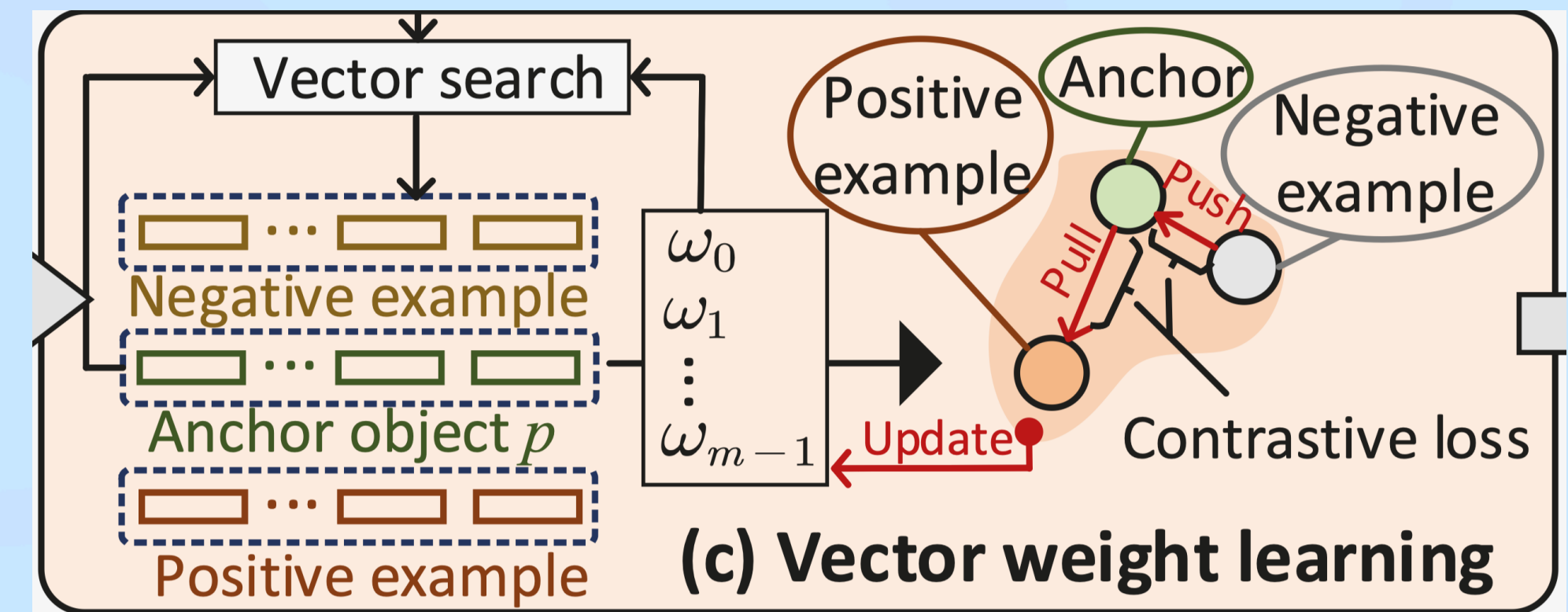
- **Loss Function.** Aims to maximize the overall similarity to positive sample, while minimize the similarity to negative samples.

$$L = \frac{1}{M} \sum_{p \in Q} -\log \frac{\exp(pp^+)}{\exp(pp^+) + \sum_{p^- \in N^-} \exp(pp^-)}$$



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Vector Weight Learning: Rethink



The weight learning process is actually a **task-specific** modal fusion process.

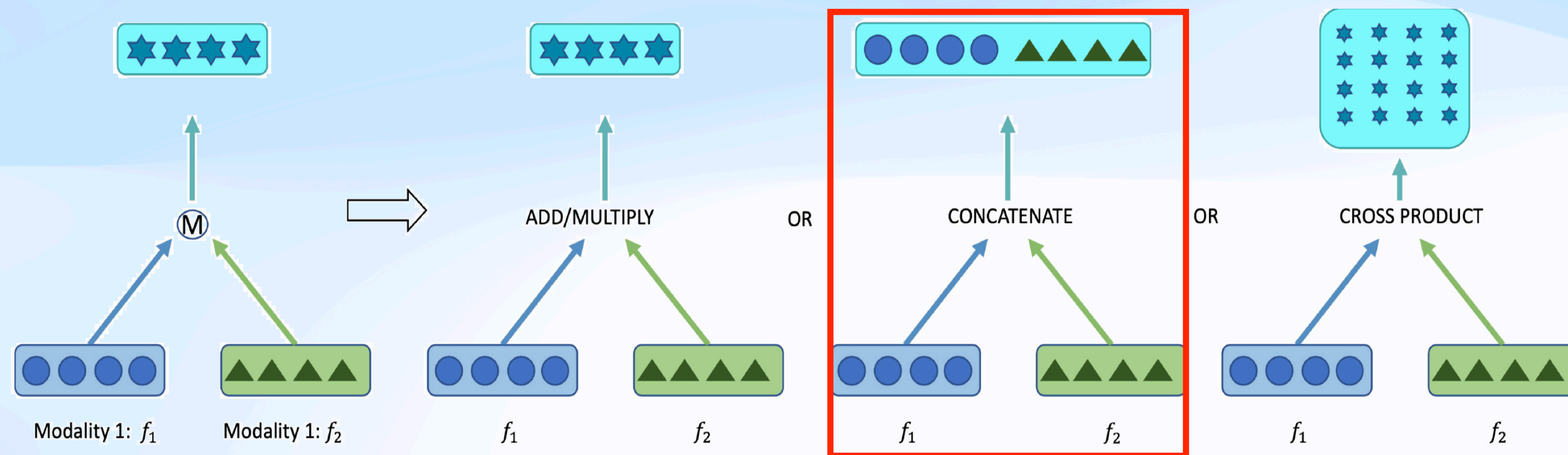
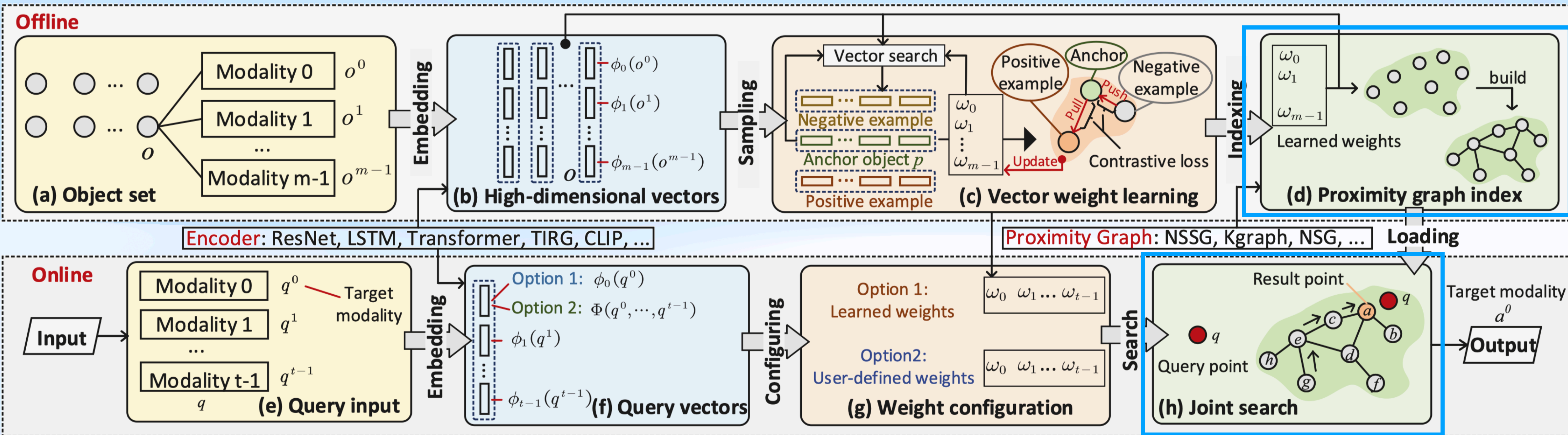


Fig. 8. The Merge operations: element-wise Addition or Multiplication, Concatenation, and Cross Product.

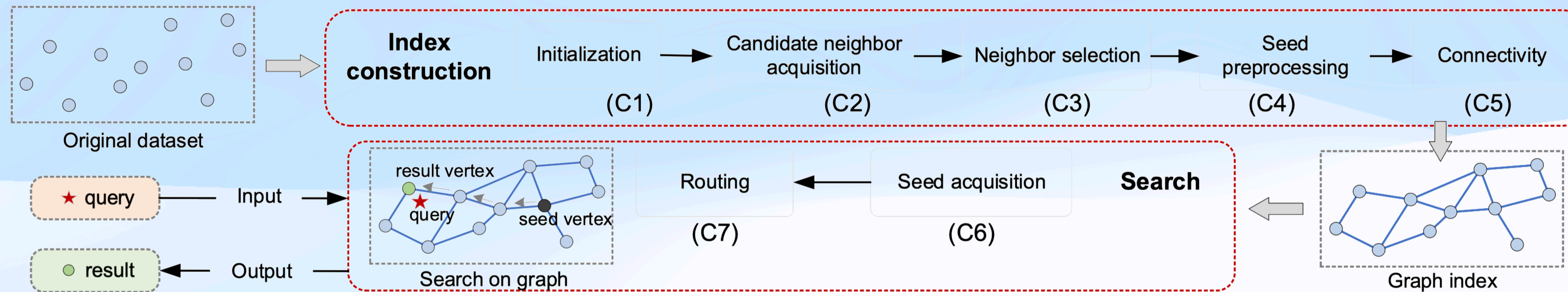
MUST Framework

Framework / ANNS



MUST Framework

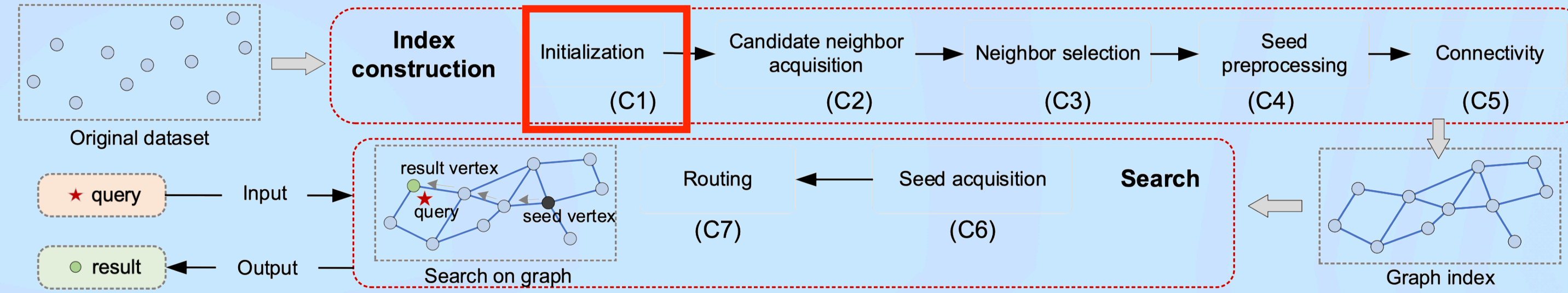
ANNS Algorithm / Graph-Based ANNS Framework



Comprehensive Survey and Experimental Comparison of Graph-Based Approximate Nearest Neighbor Search, VLDB'21

MUST Framework

ANNS / Initialization



1. Randomly selecting a set of objects $N(o)$ as neighbors for any object o .
2. Update neighbors iteratively by searching if there is a nearer object within two-jump.

Algorithm 1: CONSTRUCT FUSED INDEX

Input: Object set $\mathcal{S}$, maximum number of neighbors γ , maximum iterations ε

Output: Fused Index $G = (V, E)$ and seed vertex g

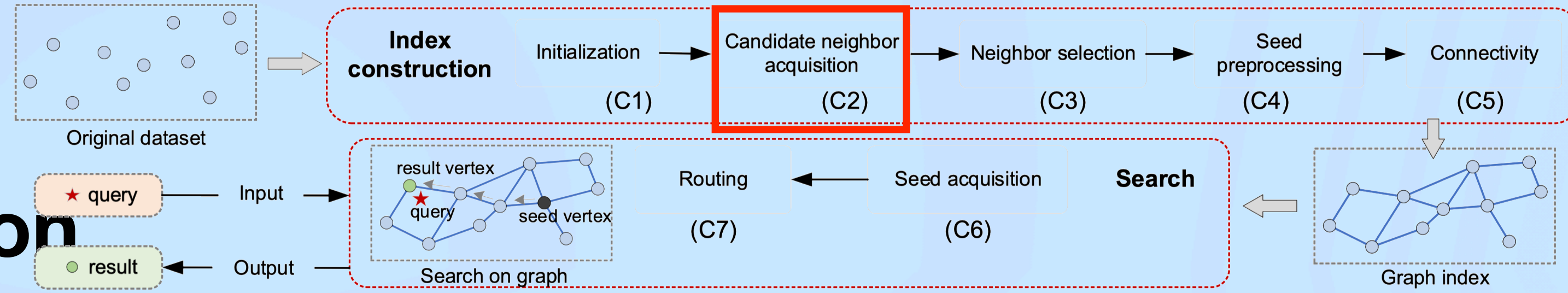
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1  $V \leftarrow \mathcal{S}; E \leftarrow \emptyset$ 
2 forall  $o \in V$  do /* ① */
3    $N(o) \leftarrow \gamma$  random objects ▷ Neighbor set
4 while  $iterations \leq \varepsilon$  do ▷ NNDescent
5   forall  $o \in V$  and  $v \in N(o)$  and  $u \in N(v) \setminus N(o)$  do
6      $z \leftarrow \arg \min_{z \in N(o)} IP(\hat{o}, \hat{z})$ 
7     if  $IP(\hat{o}, \hat{u}) > IP(\hat{o}, \hat{z})$  then
8        $N(o) \leftarrow N(o) \setminus \{z\} \cup \{u\}$ 
9 forall  $o \in V$  and  $v \in N(o)$  do /* ② */
10   $C(o) \leftarrow N(o) \cup N(v)$  ▷ Candidate neighbor
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13  while  $C(o) \neq \emptyset$  and  $|N(o)| < \gamma$  do
14     $v \leftarrow \arg \max_{v \in C(o)} IP(\hat{o}, \hat{v}); C(o) \leftarrow C(o) \setminus \{v\}$ 
15    forall  $u \in N(o)$  do ▷ MRNG strategy[25]
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18  $g \leftarrow$  nearest vertex to  $\frac{1}{|V|} \sum_{o \in V} \hat{o}$  /* ④ */
19 Ensure connectivity by BFS from  $g$  /* ⑤ */
20 return  $G = (V, E)$  and  $g$  ▷  $E = \bigcup_{o \in V} N(o)$ 

```

MUST Framework

ANNS / Candidate Acquisition



1. All nodes within two-jump are acquired as candidate neighbors.

Algorithm 1: CONSTRUCT FUSED INDEX

Input: Object set $\mathcal{S}$, maximum number of neighbors γ , maximum iterations ε

Output: Fused Index $G = (V, E)$ and seed vertex g

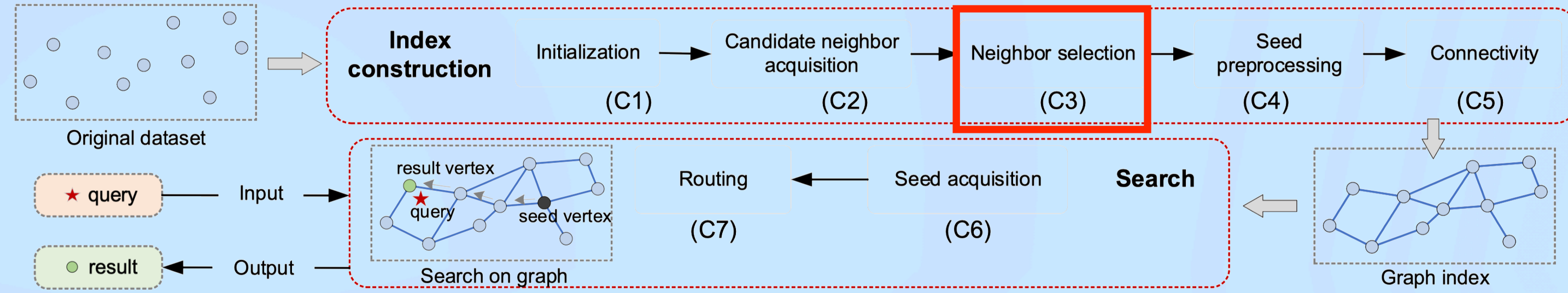
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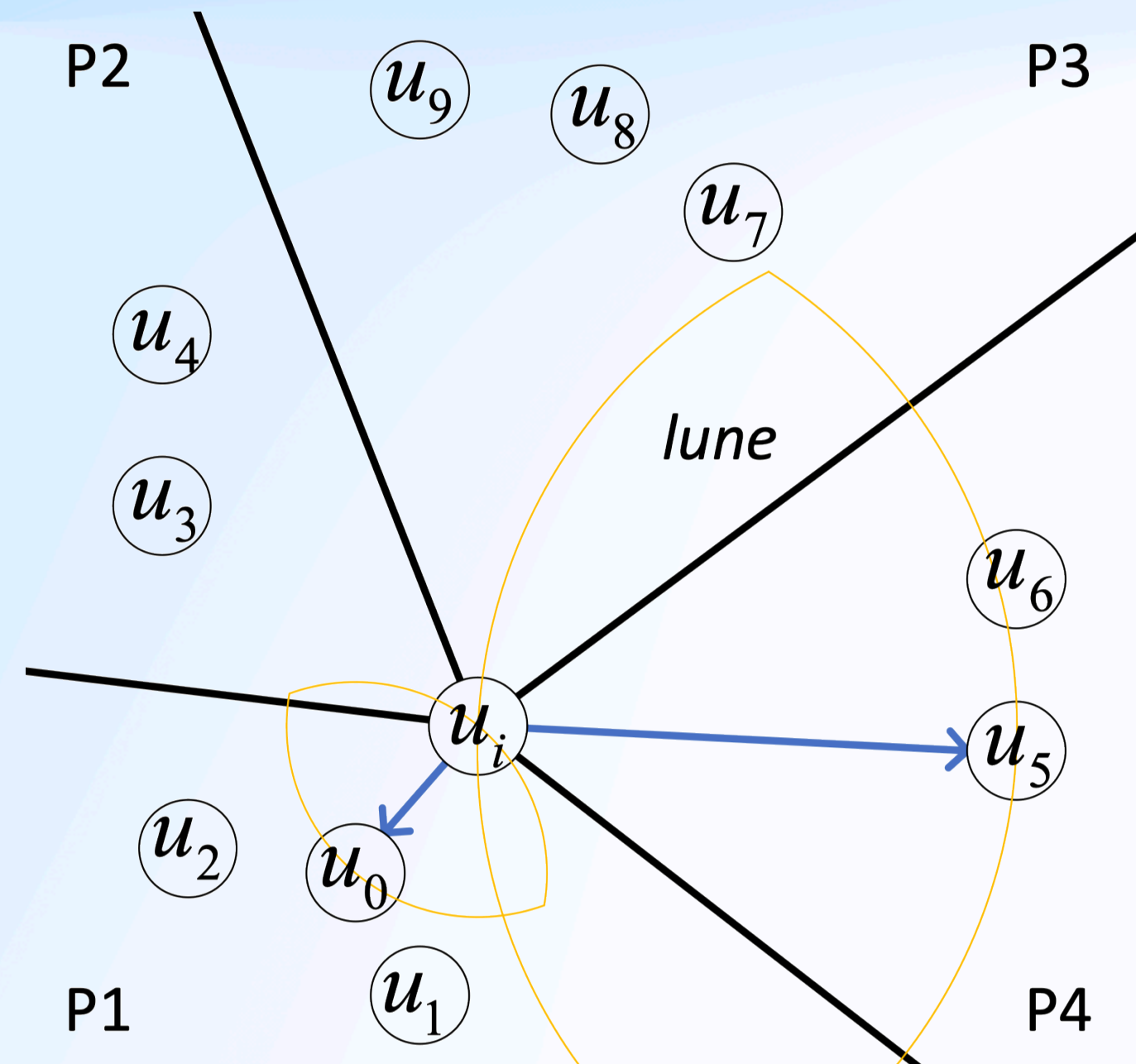
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MUST Framework

ANNS / Neighbor Selection



- **Primary objective.** Diversify the distribution of neighbors.
- **Approach (RNG).** Only include node whose **lune** does not cover the existing neighbors.
- **Lemma.** The viewing angle between two neighbors $\geq \pi/3$.



Algorithm 1: CONSTRUCT FUSED INDEX

Input: Object set $\mathcal{S}$, maximum number of neighbors γ , maximum iterations ε

Output: Fused Index $G = (V, E)$ and seed vertex g

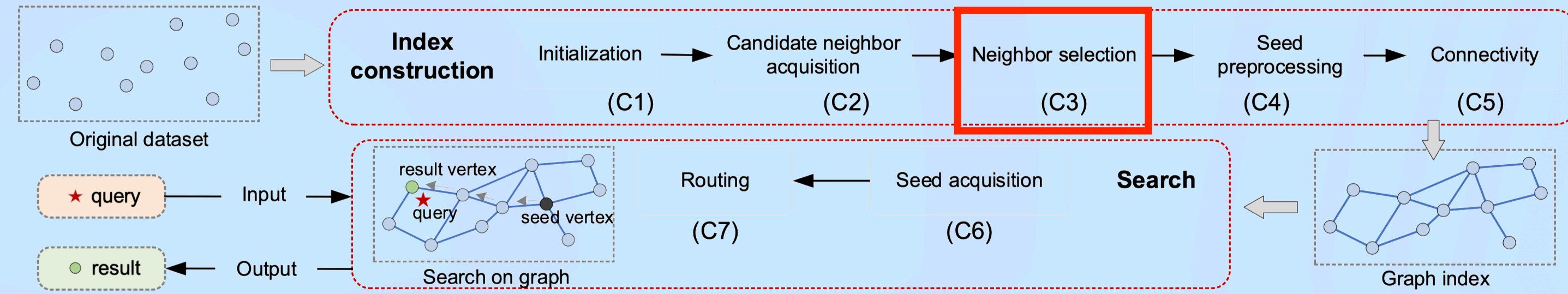
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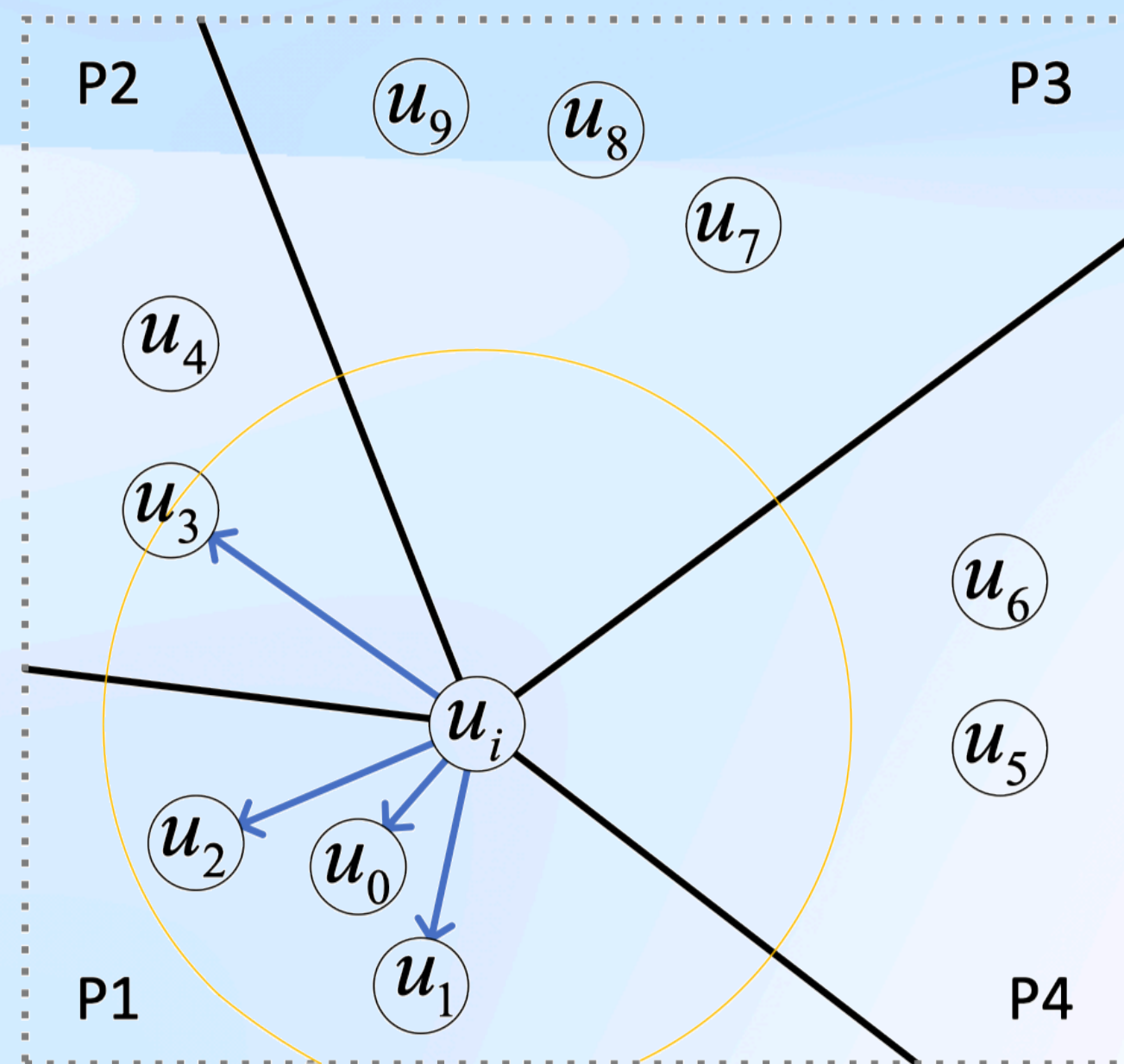
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MUST Framework

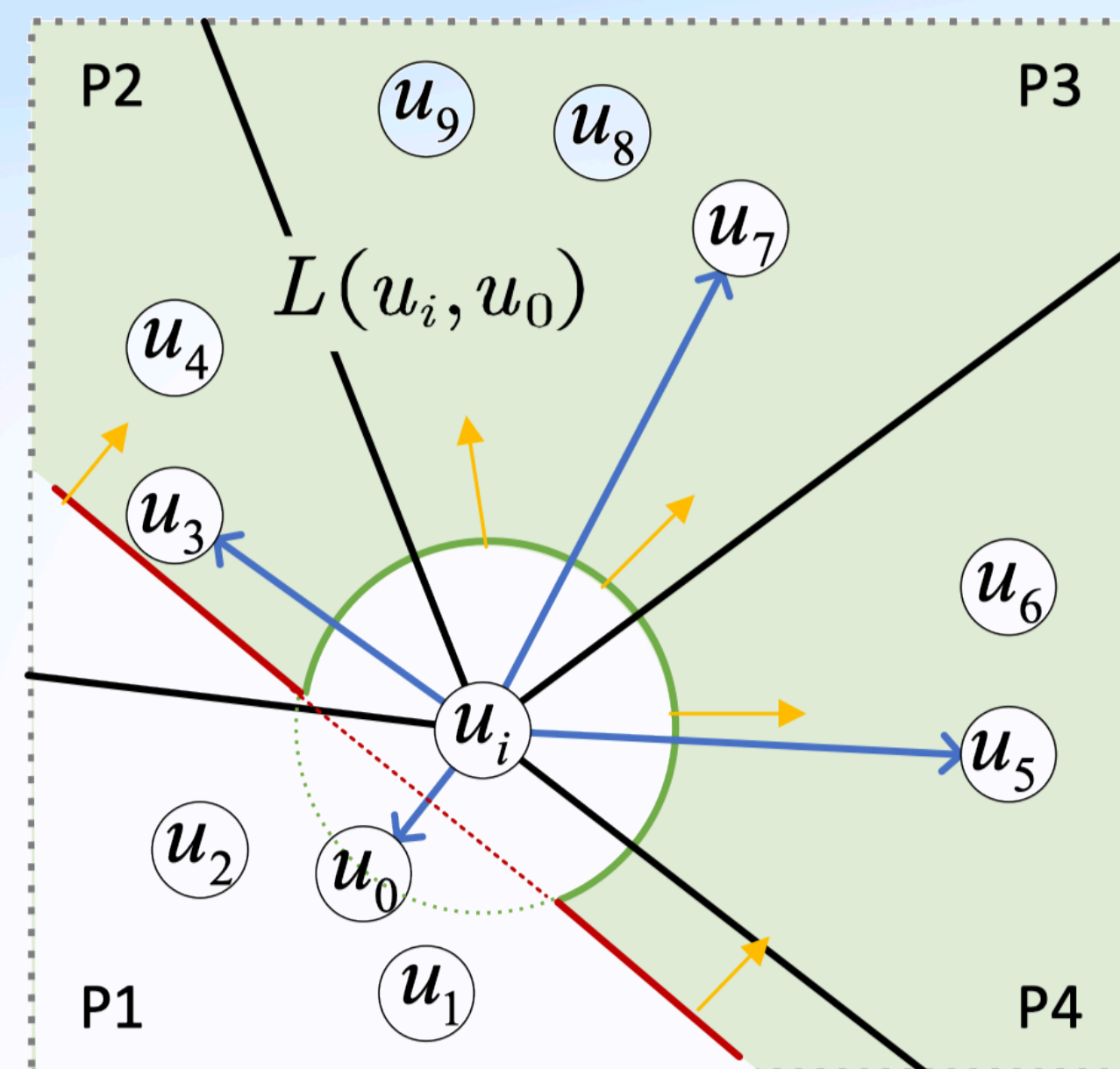
ANNS / Neighbor Selection



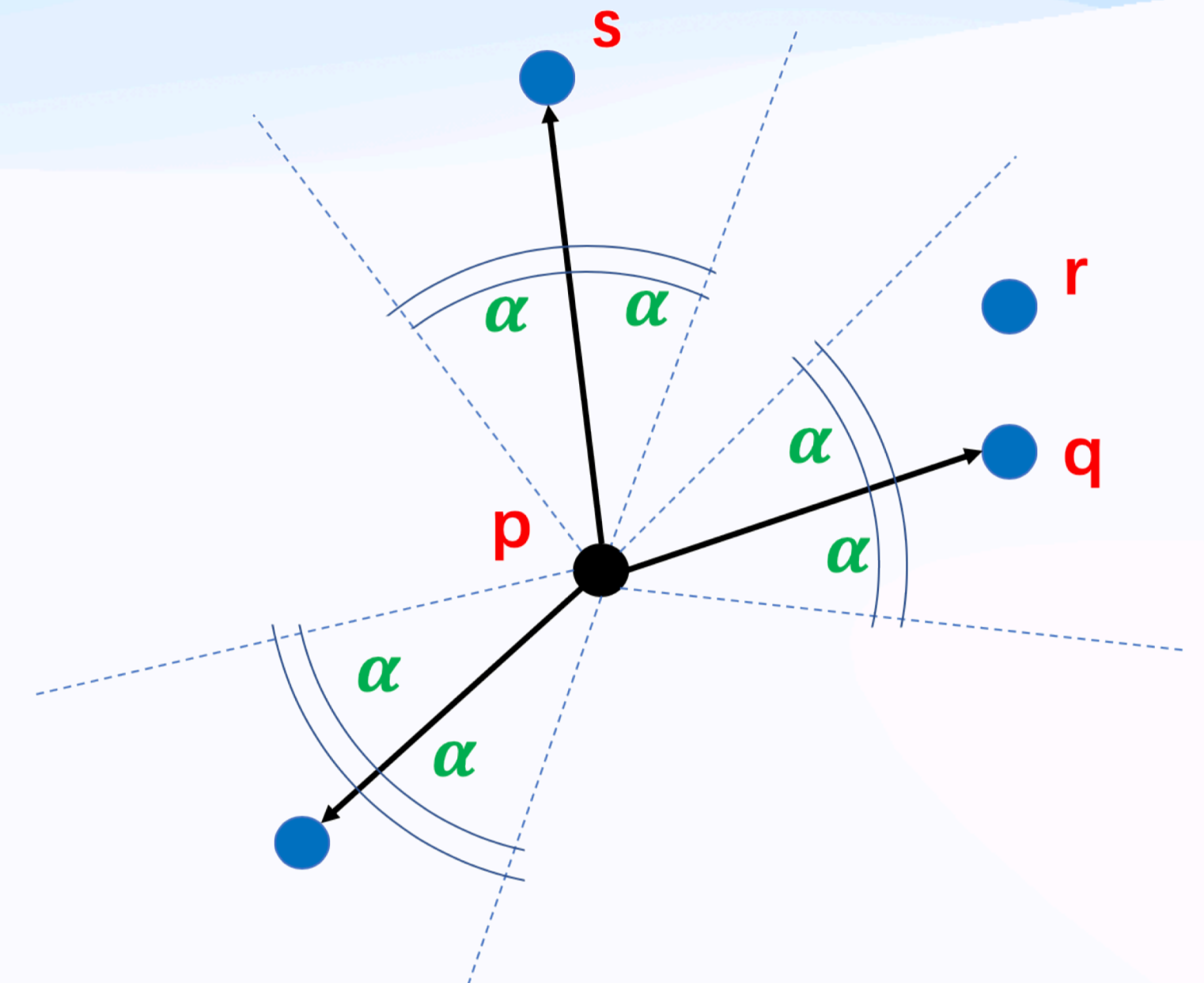
- **Primary objective.** Diversify the distribution of neighbors?
- **Approaches?**



Naive Edge Selection



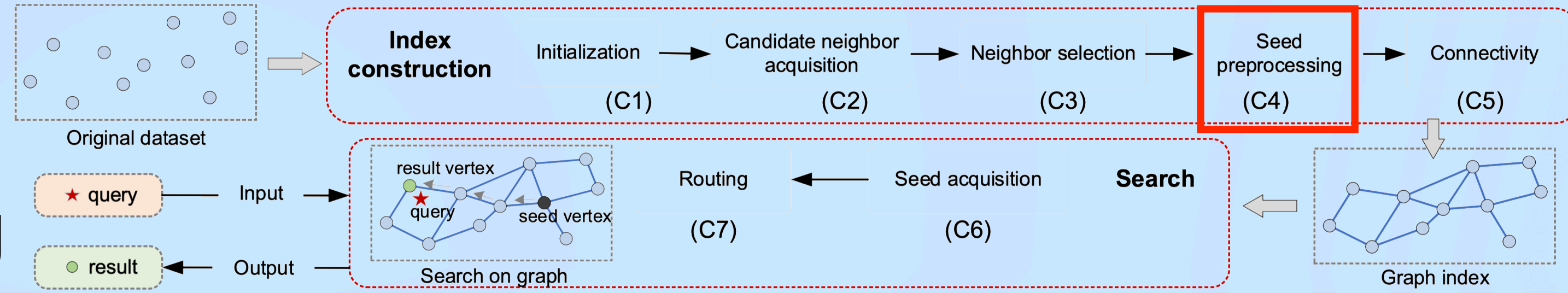
An Efficient and Robust Framework for Approximate Nearest Neighbor Search with Attribute Constraint, NeurIPS'23



High Dimensional Similarity Search with Satellite System Graph: Efficiency, Scalability, and Unindexed Query Compatibility, TAPMI'20

MUST Framework

ANNS / Seed Preprocessing



- Use a **fixed** seed as start vertex for all searches.

1. Compute the centroid of all vertexes.
2. Find the vertex nearest to centroid.

Algorithm 1: CONSTRUCT FUSED INDEX

Input: Object set $\mathcal{S}$, maximum number of neighbors γ , maximum iterations ε

Output: Fused Index $G = (V, E)$ and seed vertex g

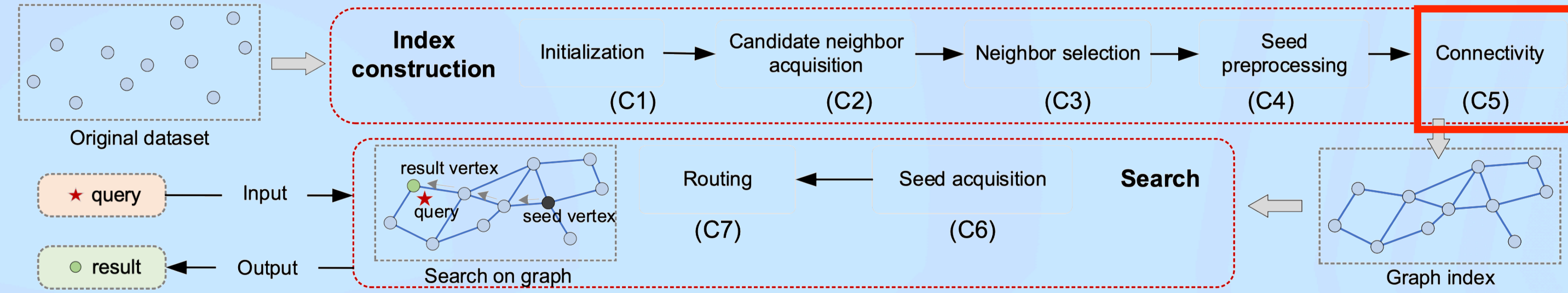
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20 return  $G = (V, E)$  and  $g$  ▷  $E = \bigcup_{o \in V} N(o)$ 

```

MUST Framework

ANNS / Connectivity



1. Do BFS starting from seed.
2. If any unvisited vertexes, connections are manually constructed between visited ones and unvisited ones

Algorithm 1: CONSTRUCT FUSED INDEX

Input: Object set $\mathcal{S}$, maximum number of neighbors γ , maximum iterations ε

Output: Fused Index $G = (V, E)$ and seed vertex g

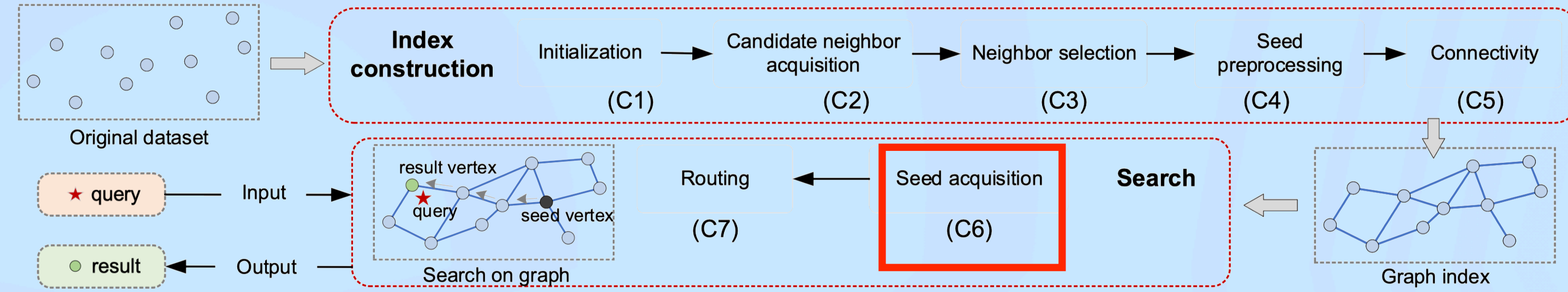
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```


MUST Framework

ANNS / Seed Acquisition



1. Initialization Result set R with seed g .
2. Set visited vertices set H as empty.
3. Add another $l - 1$ random vertices into initialization result set R .

Algorithm 2: JOINT SEARCH

Input: Fused index $G = (V, E)$, multimodal query q , seed vertex g , number of results k , result set size $l (> k)$

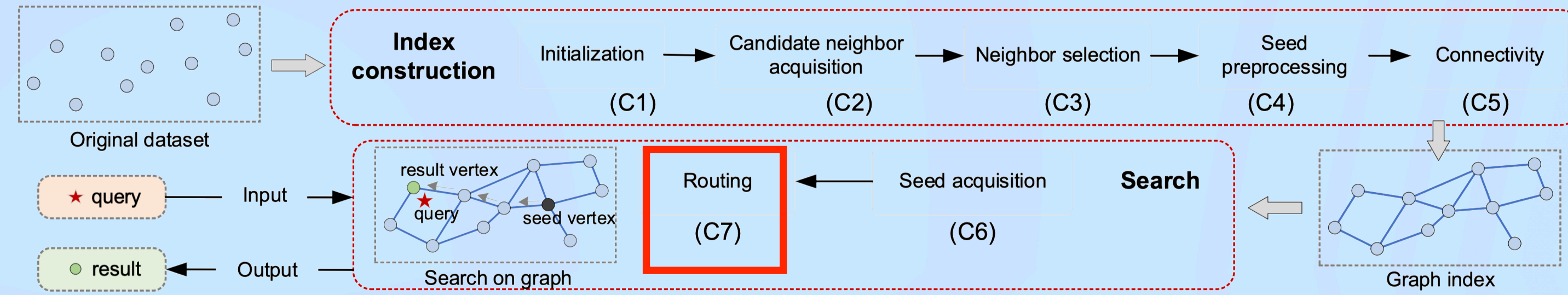
Output: approximate top- k results of q

```

1  $R \leftarrow \{g\}; H \leftarrow \emptyset$ 
2  $C \leftarrow l - 1$  random vertices
3  $R \leftarrow R \cup C$  ▷ sorted by IP to  $q$ 
4 while  $(R \setminus H) \neq \emptyset$  do ▷ unvisited vertices
5      $v \leftarrow$  unvisited nearest vertex to  $q$  in  $R$  ▷  $v \notin H$ 
6      $H \leftarrow H \cup \{v\}$  ▷ mark  $v$  as visited
7     forall  $u \in N(v)$  and  $u \notin H$  do
8          $z \leftarrow \arg \min_{z \in R} IP(\hat{q}, \hat{z})$ 
9         if  $IP(\hat{q}, \hat{z}) < IP(\hat{q}, \hat{u})$  then
10              $R \leftarrow R \setminus \{z\} \cup \{u\}$  ▷ update  $R$ 
11 return top- $k$  nearest vertices in  $R$ 
    
```

MUST Framework

ANNS / Routing



- Greedy Routing.
- **Optimization:** Early exit in similarity comparison.

$$v_1 = \left(\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3} \right)$$

$$v_2 = \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 0 \right)$$

$$v_3 = (0, 0, 1)$$

Algorithm 2: JOINT SEARCH

Input: Fused index $G = (V, E)$, multimodal query q , seed vertex g , number of results k , result set size l ($> k$)

Output: approximate top- k results of q

```

1  $R \leftarrow \{g\}; H \leftarrow \emptyset$ 
2  $C \leftarrow l - 1$  random vertices
3  $R \leftarrow R \cup C$   $\triangleright$  sorted by IP to  $q$ 
4 while  $(R \setminus H) \neq \emptyset$  do  $\triangleright$  unvisited vertices
5    $v \leftarrow$  unvisited nearest vertex to  $q$  in  $R$   $\triangleright v \notin H$ 
6    $H \leftarrow H \cup \{v\}$   $\triangleright$  mark  $v$  as visited
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11 return top- $k$  nearest vertices in  $R$ 
  
```


Evaluation

Evaluation

Dataset

Dataset	# Modality	# Object	# Query	Type	Source
CelebA [33]	2	191,549	34,326	Image [*] , Text	real-world
MIT-States [41]	2	53,743	72,732	Image [*] , Text	real-world
Shopping [42]	2	96,009	47,658	Image [*] , Text	real-world
MS-COCO [43]	3	19,711	1237	Image [*] $\times$ 2, Text	real-world
CelebA+ [33]	4	191,549	34,326	Image [*] $\times$ 3, Text	real-world
ImageText1M [44]	2	1,000,000	10,000	Image [*] , Text	semi-synthetic
AudioText1M [45]	2	992,272	200	Audio [*] , Text	semi-synthetic
VideoText1M [46]	2	1,000,000	10,000	Video [*] , Text	semi-synthetic
ImageText16M [47]	2	16,000,000	10,000	Image [*] , Text	semi-synthetic

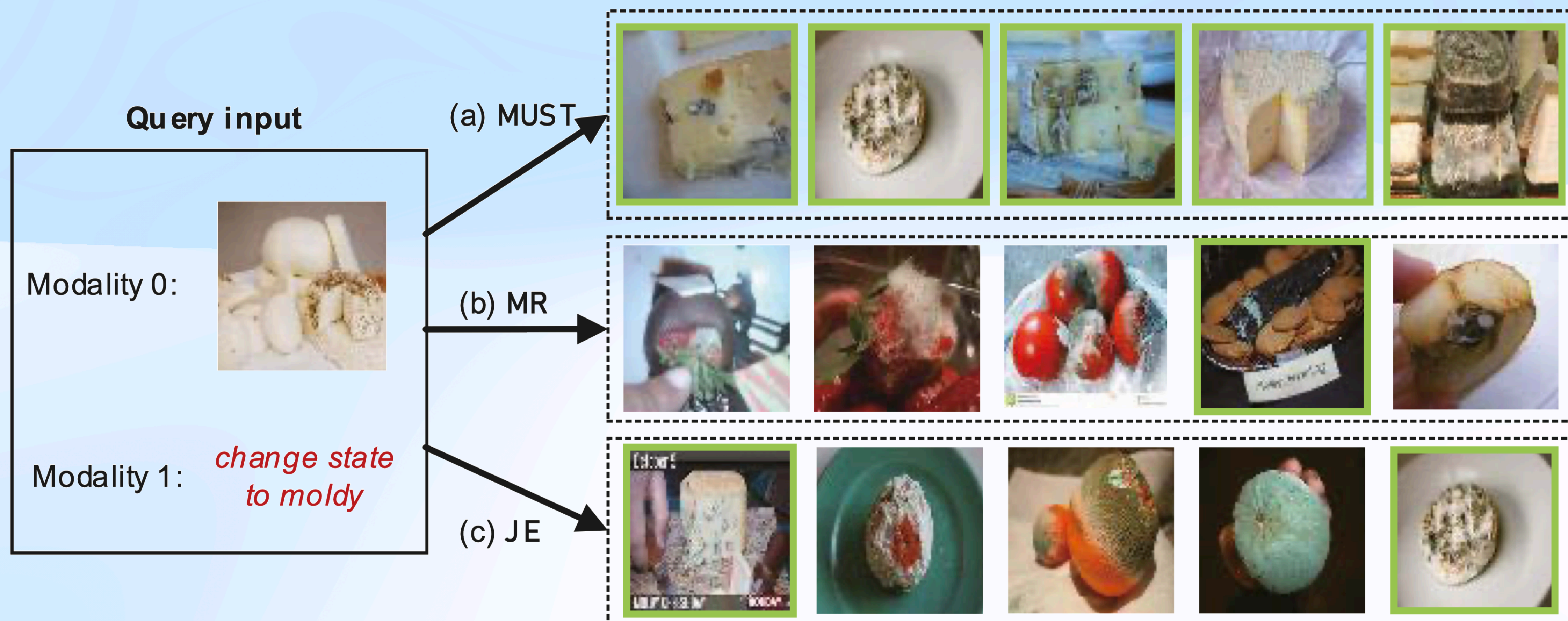
Evaluation

Accuracy

Framework	Encoder	<i>Recall@1(1)</i>	<i>Recall@5(1)</i>	<i>Recall@10(1)</i>	<i>SME</i>
JE	TIRG	0.1181	0.3027	0.4175	0.1574
	CLIP	0.2236	0.4979	0.6187	0.1382
MR	ResNet17+LSTM	0.3998	0.6336	0.7106	0.1222
	ResNet50+LSTM	0.5401	0.7104	0.7639	0.1012
	ResNet17+Transformer	0.2435	0.4110	0.4931	0.1381
	ResNet50+Transformer	0.3112	0.4475	0.5142	0.1404
	TIRG+LSTM	0.3768	0.6574	0.7691	0.1283
	TIRG+Transformer	0.2830	0.4918	0.5834	0.1395
	CLIP+LSTM	0.4911	0.7619	0.8436	0.1108
	CLIP+Transformer	0.3707	0.5912	0.6751	0.1285
MUST (ours)	ResNet17+LSTM	0.5275	0.7897	0.8780	0.0915
	ResNet50+LSTM	0.6655(↑23.2%)	0.8558(↑12.3%)	0.9127(↑8.2%)	0.0738
	ResNet17+Transformer	0.3325	0.4828	0.5548	0.1272
	ResNet50+Transformer	0.3743	0.4866	0.5367	0.1344
	TIRG+LSTM	0.4202	0.7012	0.8137	0.1184
	TIRG+Transformer	0.3131	0.4800	0.5543	0.1333
	CLIP+LSTM	0.5376	0.7859	0.8678	0.1006
	CLIP+Transformer	0.4190	0.5262	0.5731	0.1229

Evaluation

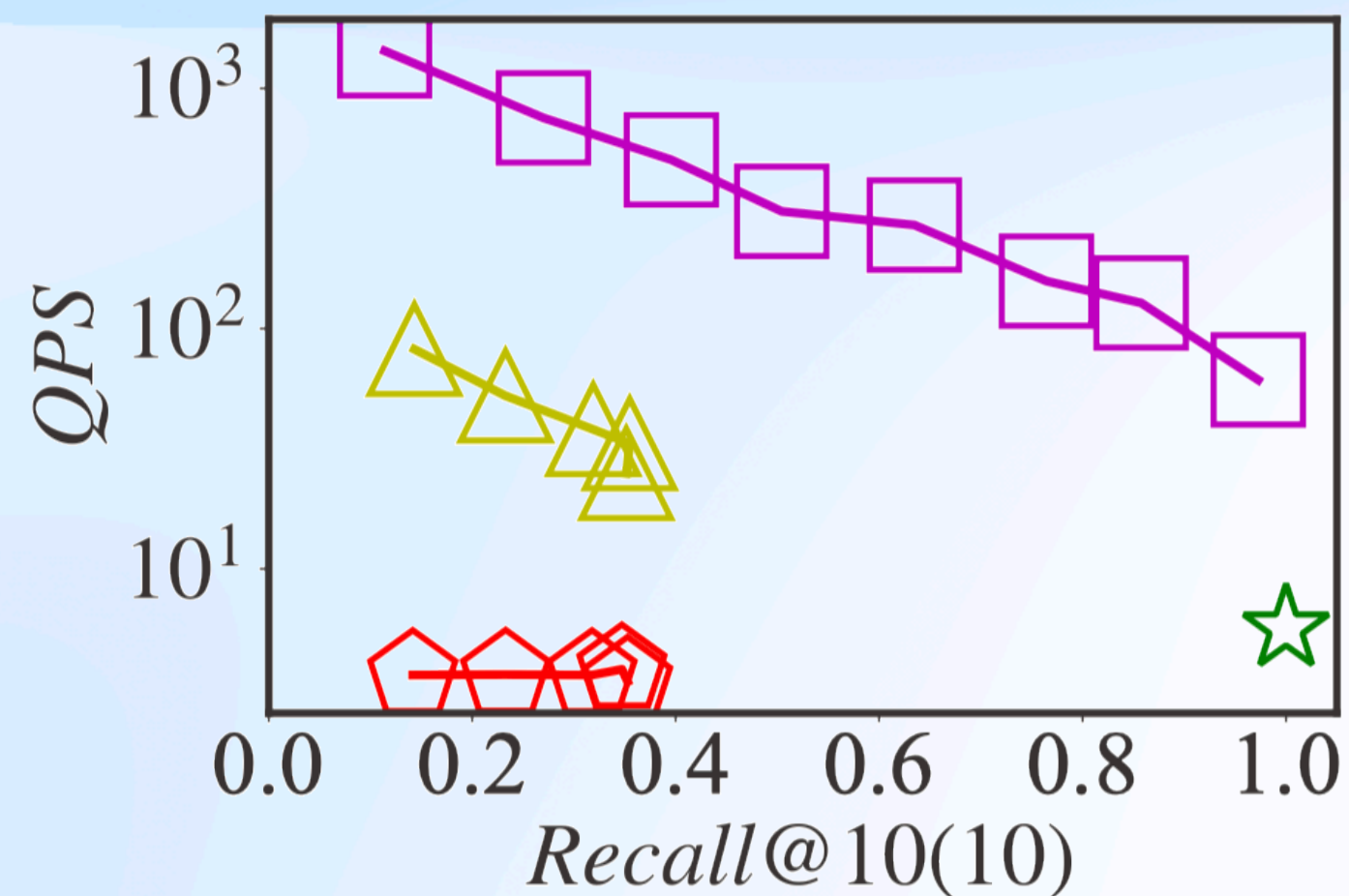
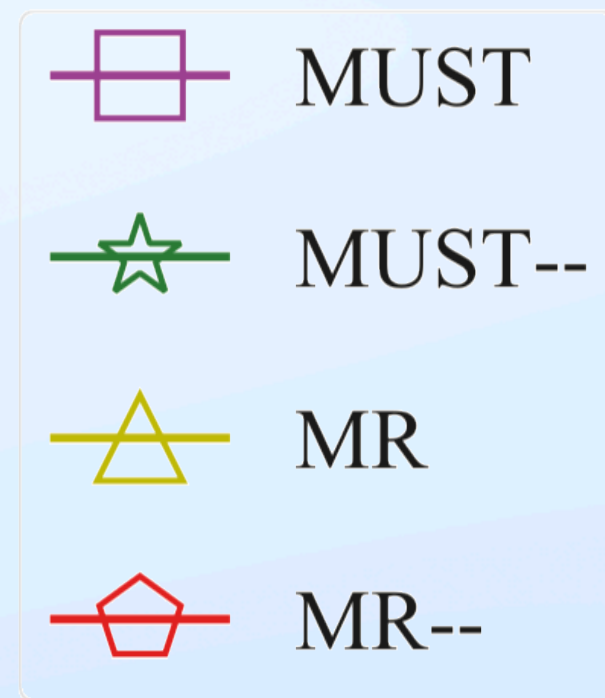
MIT-States Dataset Query Example



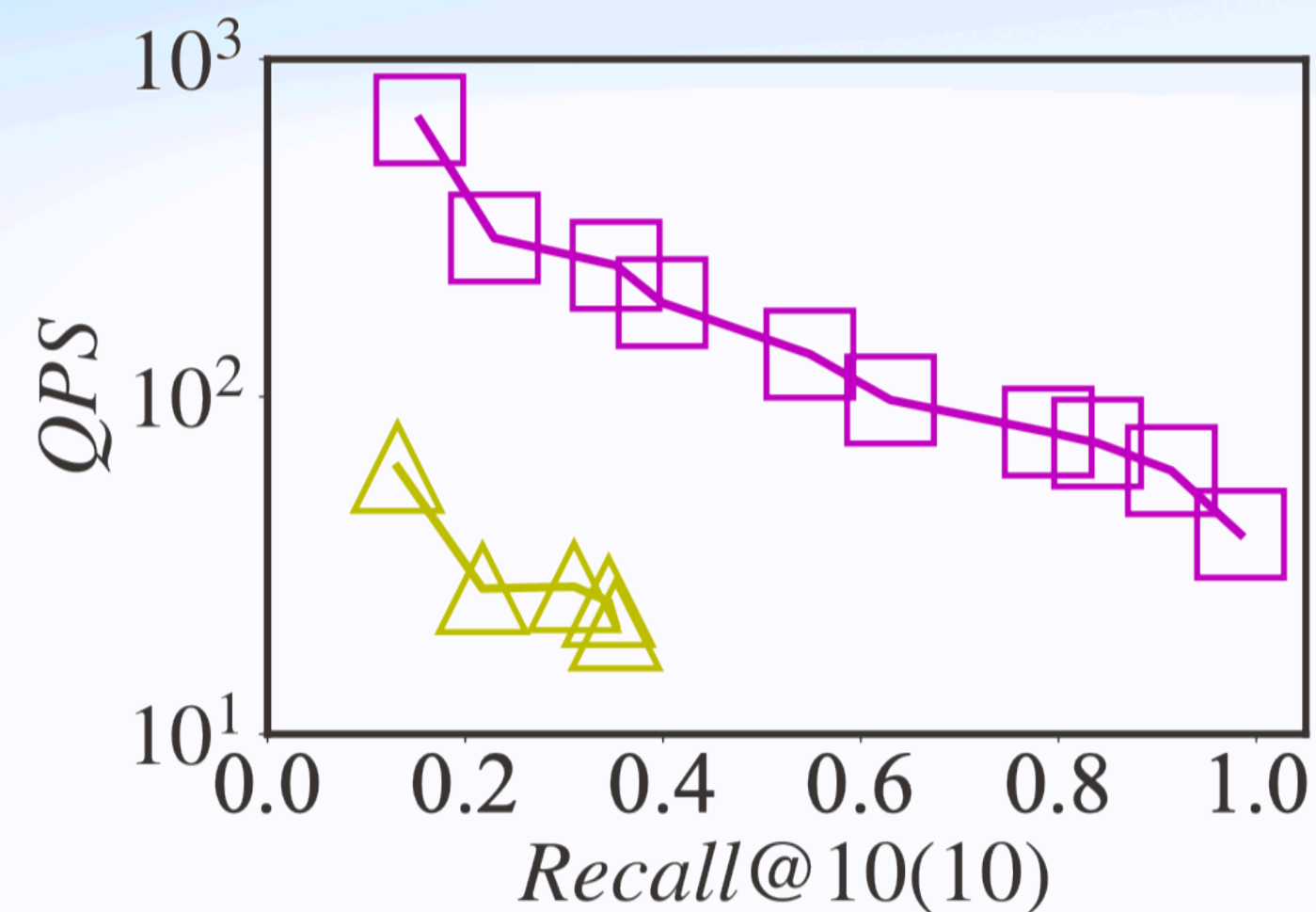
Evaluation

Efficiency (QPS-Recall@10)

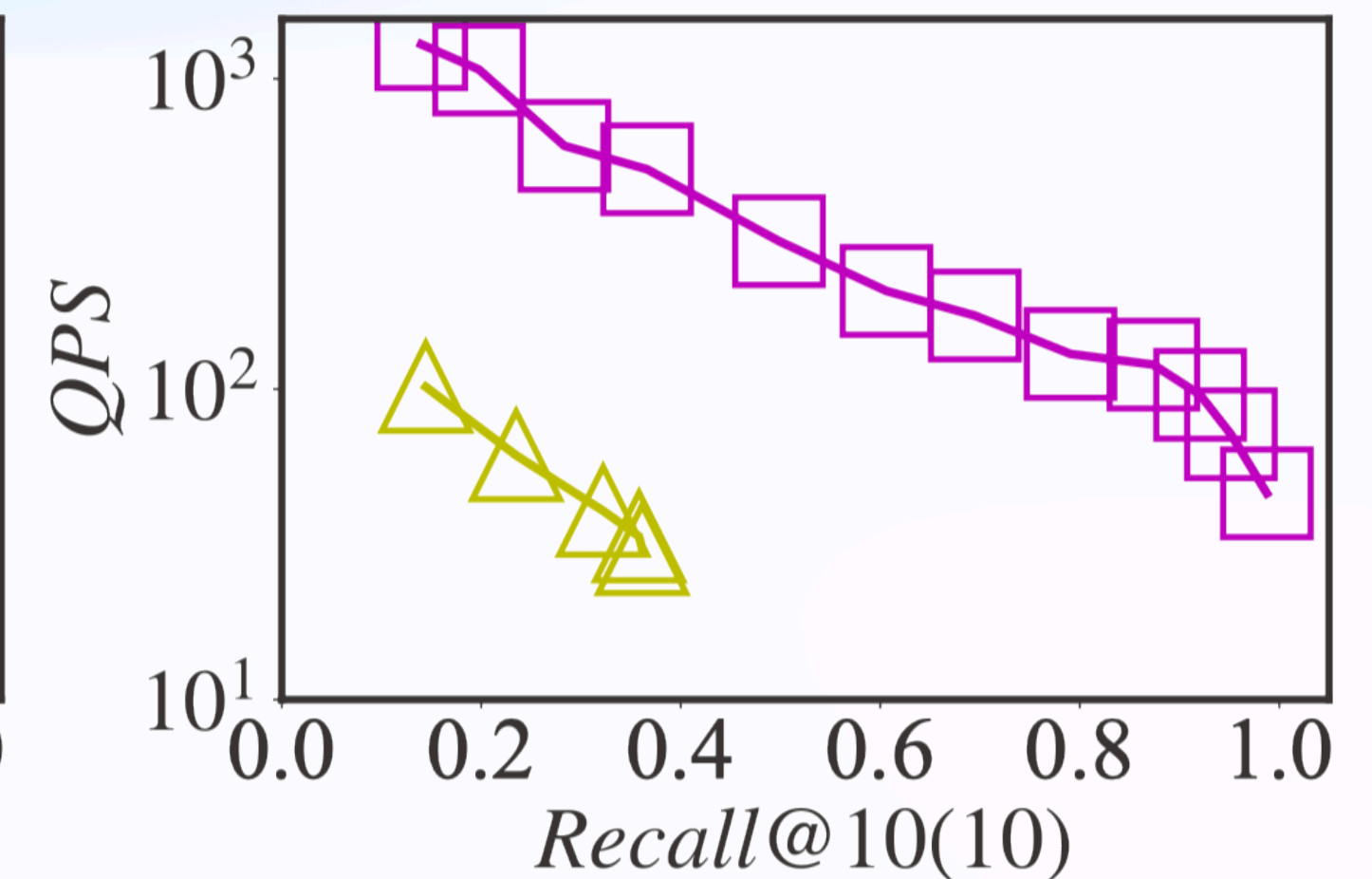
- MUST—/MR—: Brute-force vector search version.
- MUST outperforms MR in QPS-Recall by 10x.



(a) ImageText1M



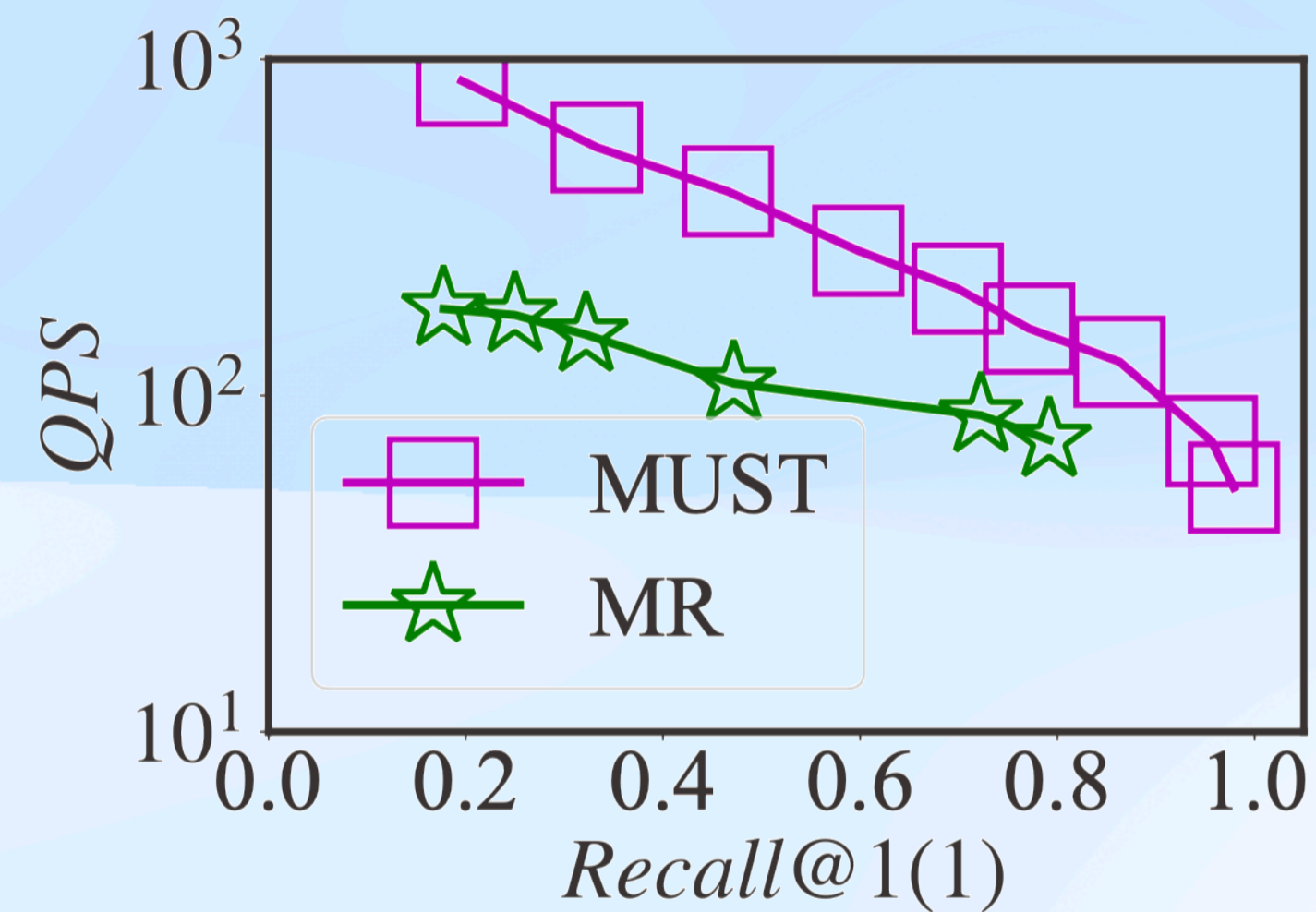
(b) AudioText1M



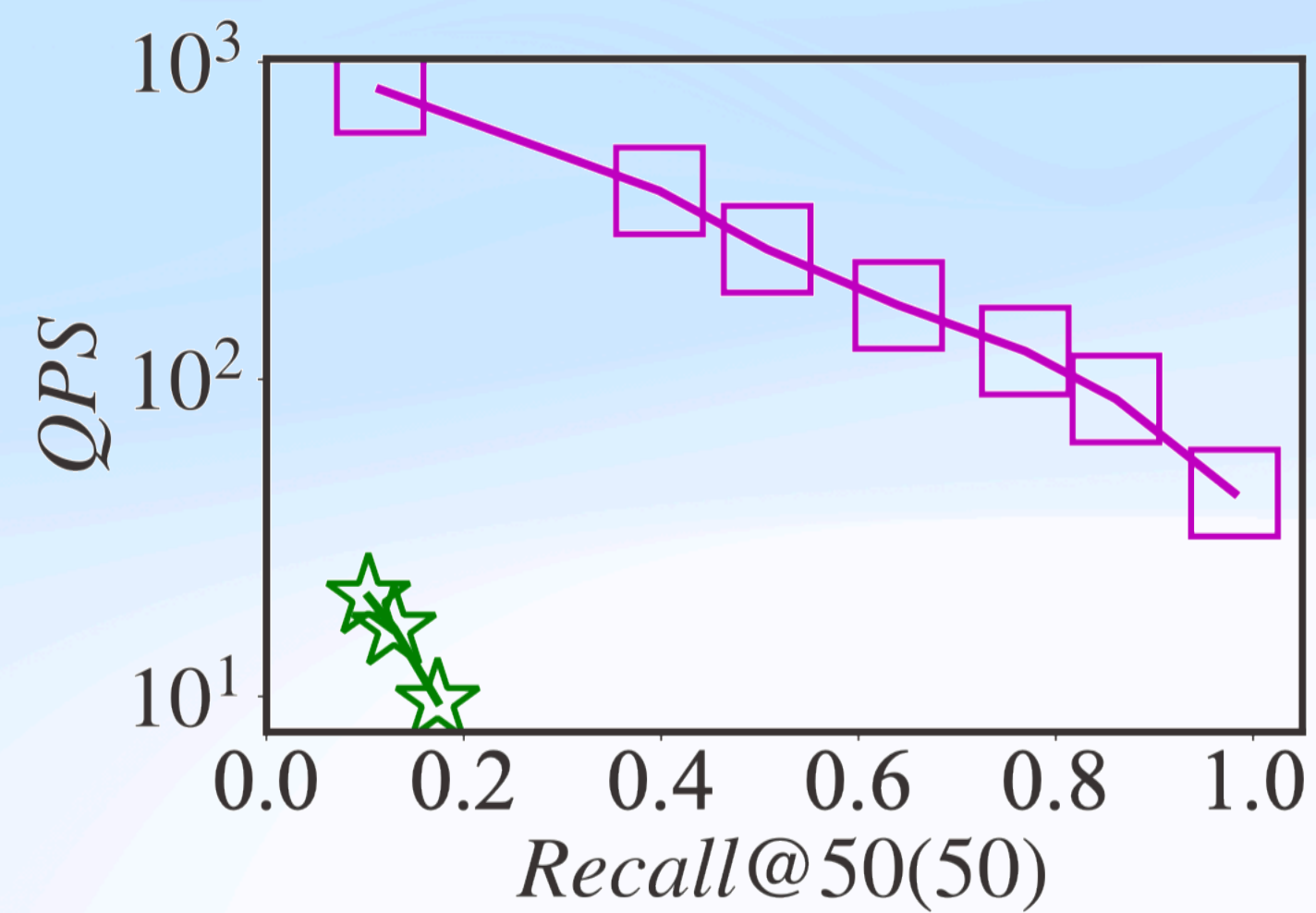
(c) VideoText1M

Evaluation

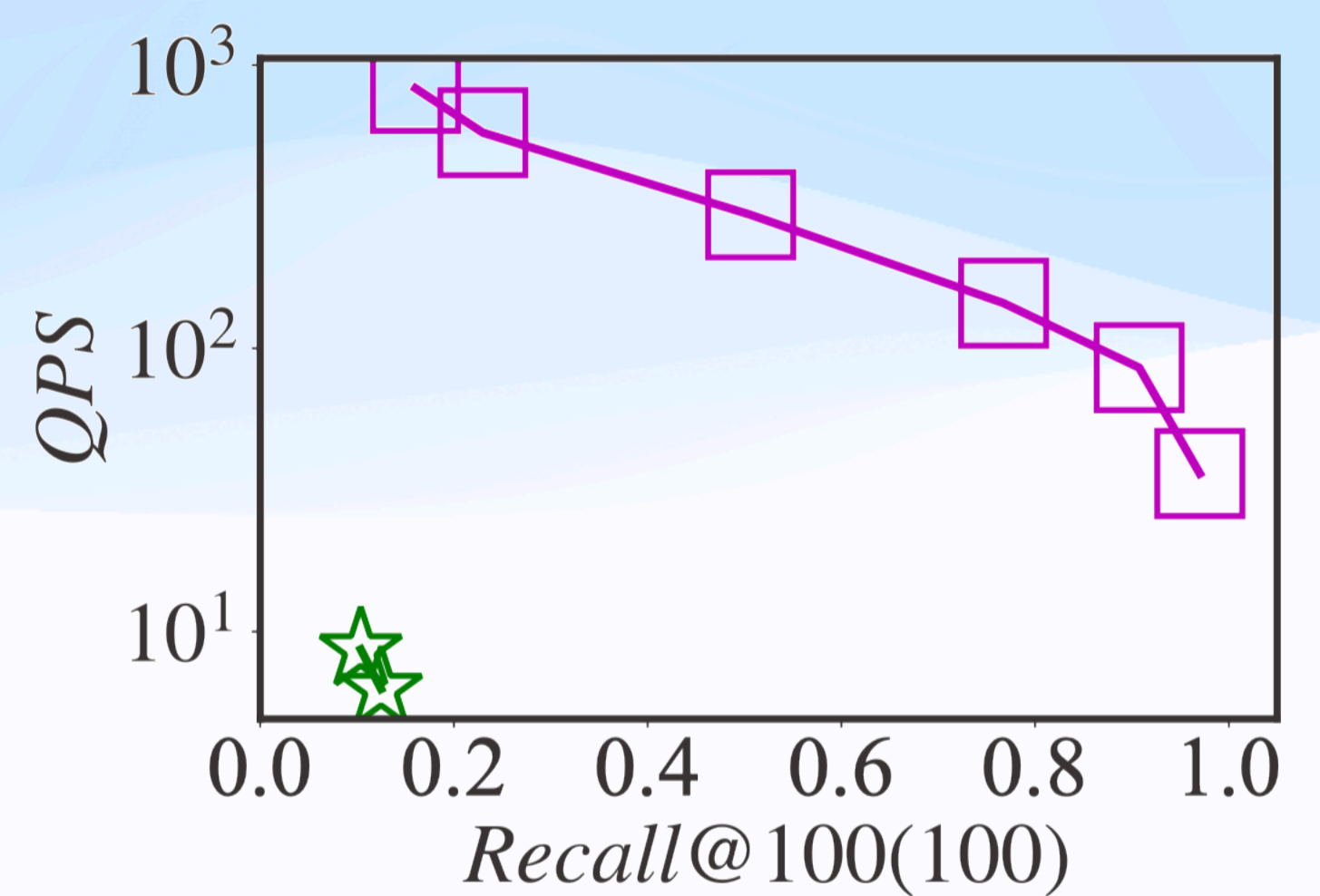
QPS-Recall@N as N increases



(a) $k = 1$



(b) $k = 50$

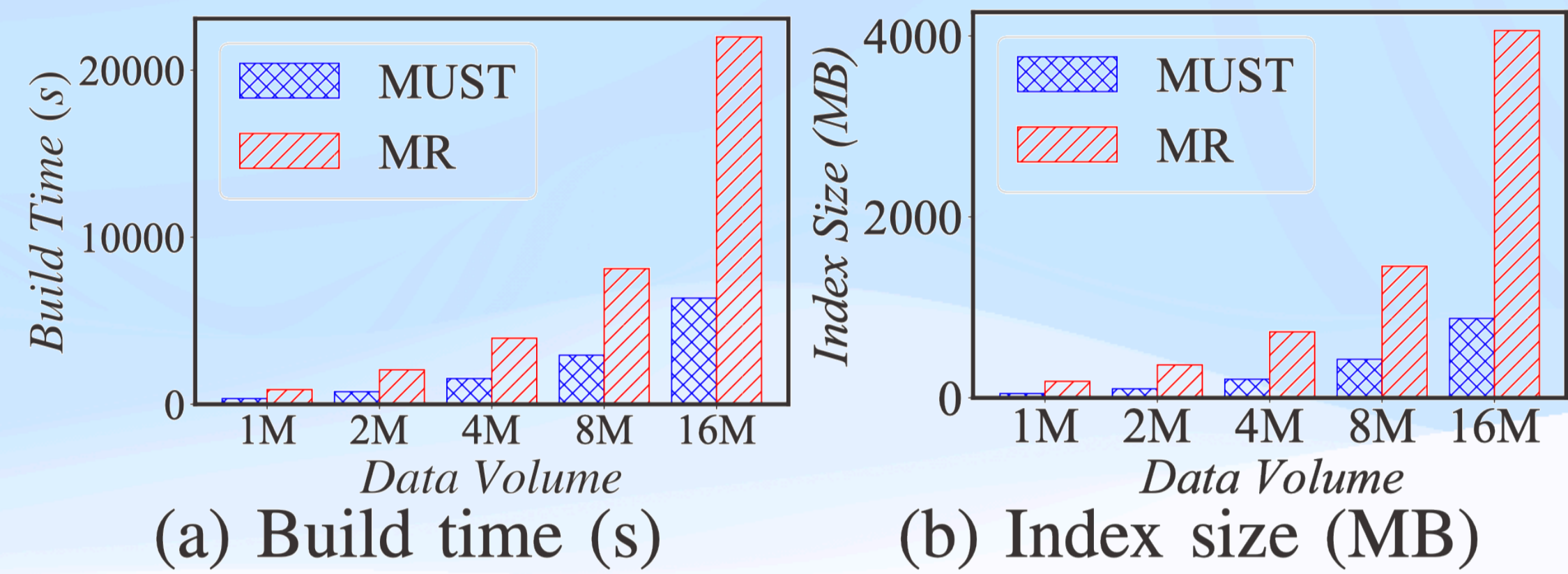


(c) $k = 100$

Evaluation

Scalability

- **Data Volume**
 - Index Building Time and Index Size
 - Query Response Time



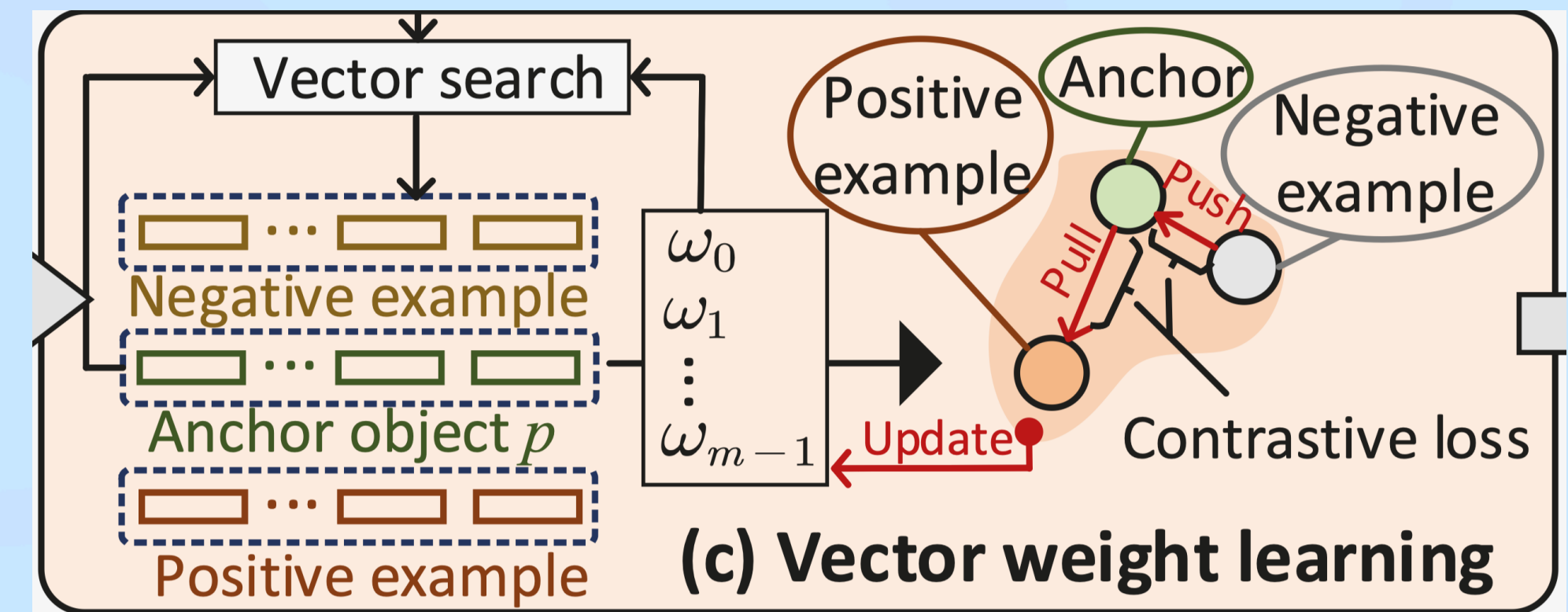
Scale	1M	2M	4M	8M	16M
MUST--	15.4	32.8	67.5	129.9	266.9
MUST	2.7 (↓82.5%)	2.7 (↓91.8%)	3.4 (↓95.0%)	3.4 (↓97.4%)	4.4 (↓98.4%)

- **Number of Modalities**

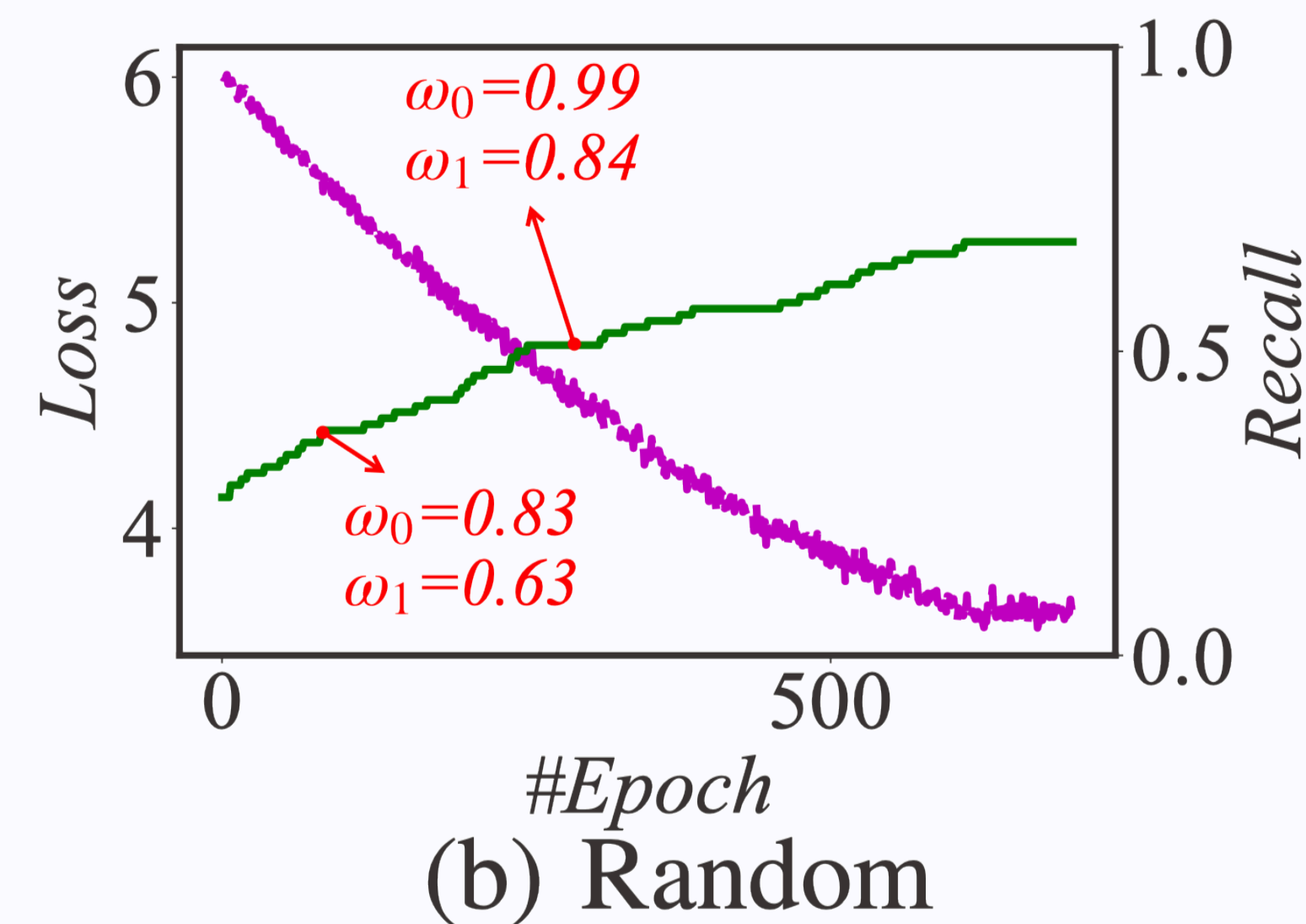
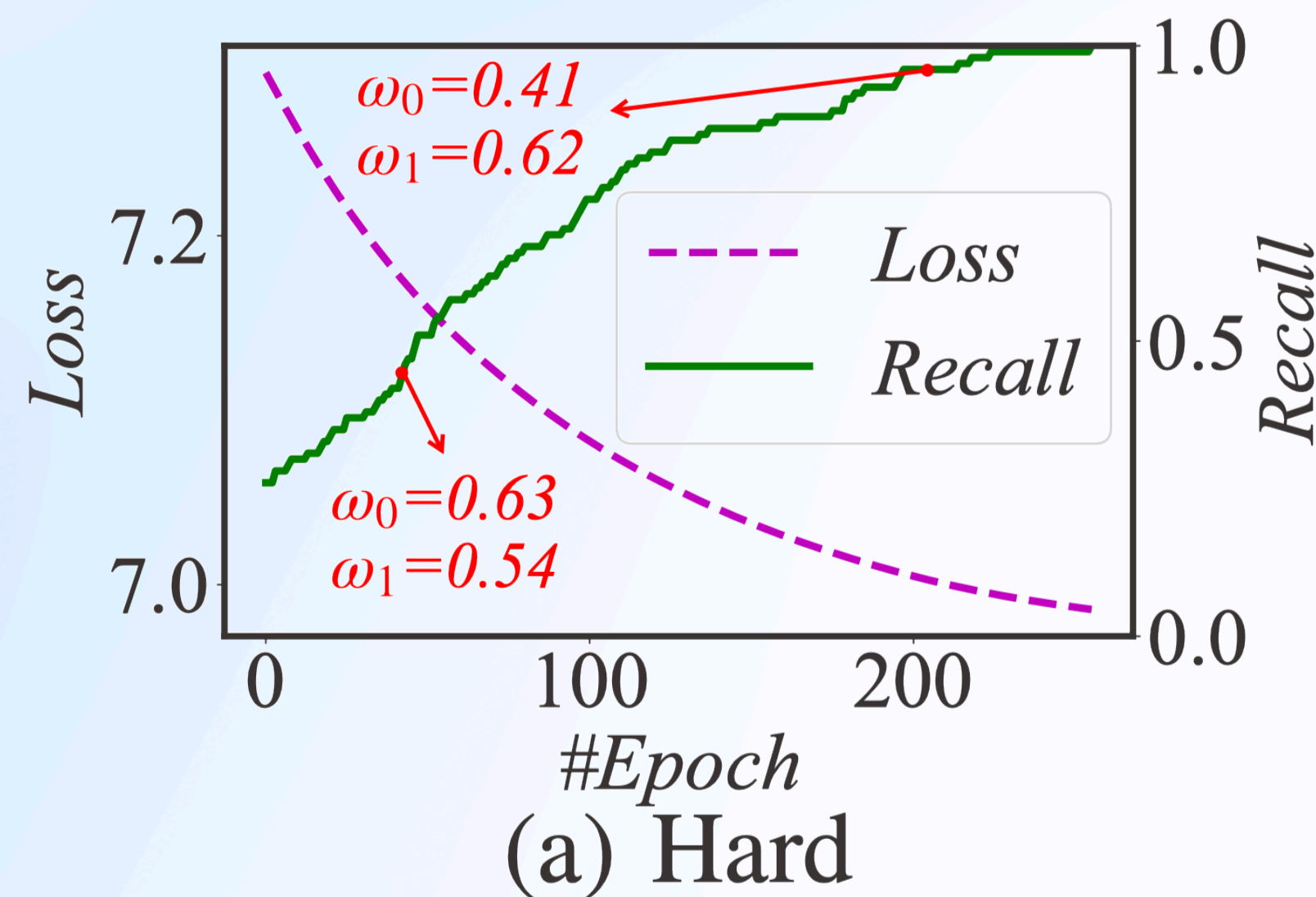
# Modality (m)	2	3	4
MR ($Recall@1(1)$)	0.4578	0.4613	0.4599
MUST ($Recall@1(1)$)	0.6388	0.6771	0.6956

Evaluation

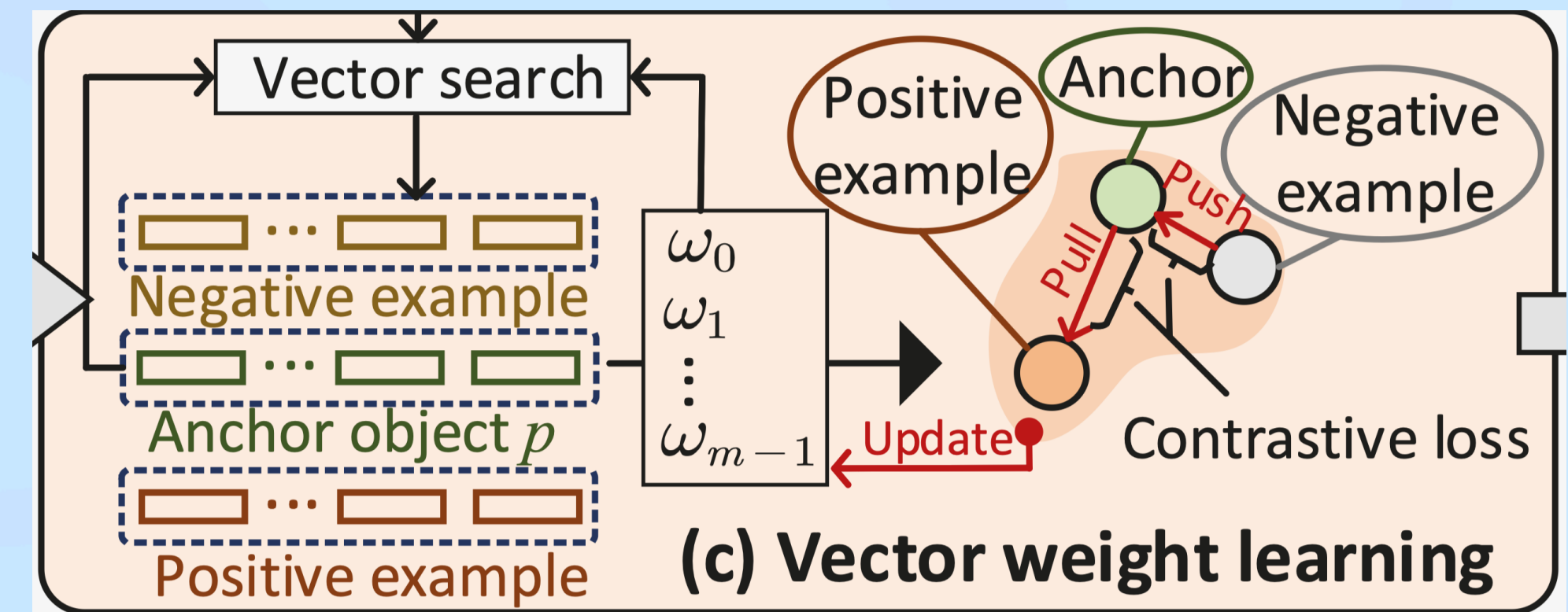
Vector Weight Learning



- Compared to random negative samples, hard negative samples (TopK samples, which are easy to be confused with the ground truth during search) can:
 - Converge Faster
 - Reach Higher Recall Rate



User-Define Weight?



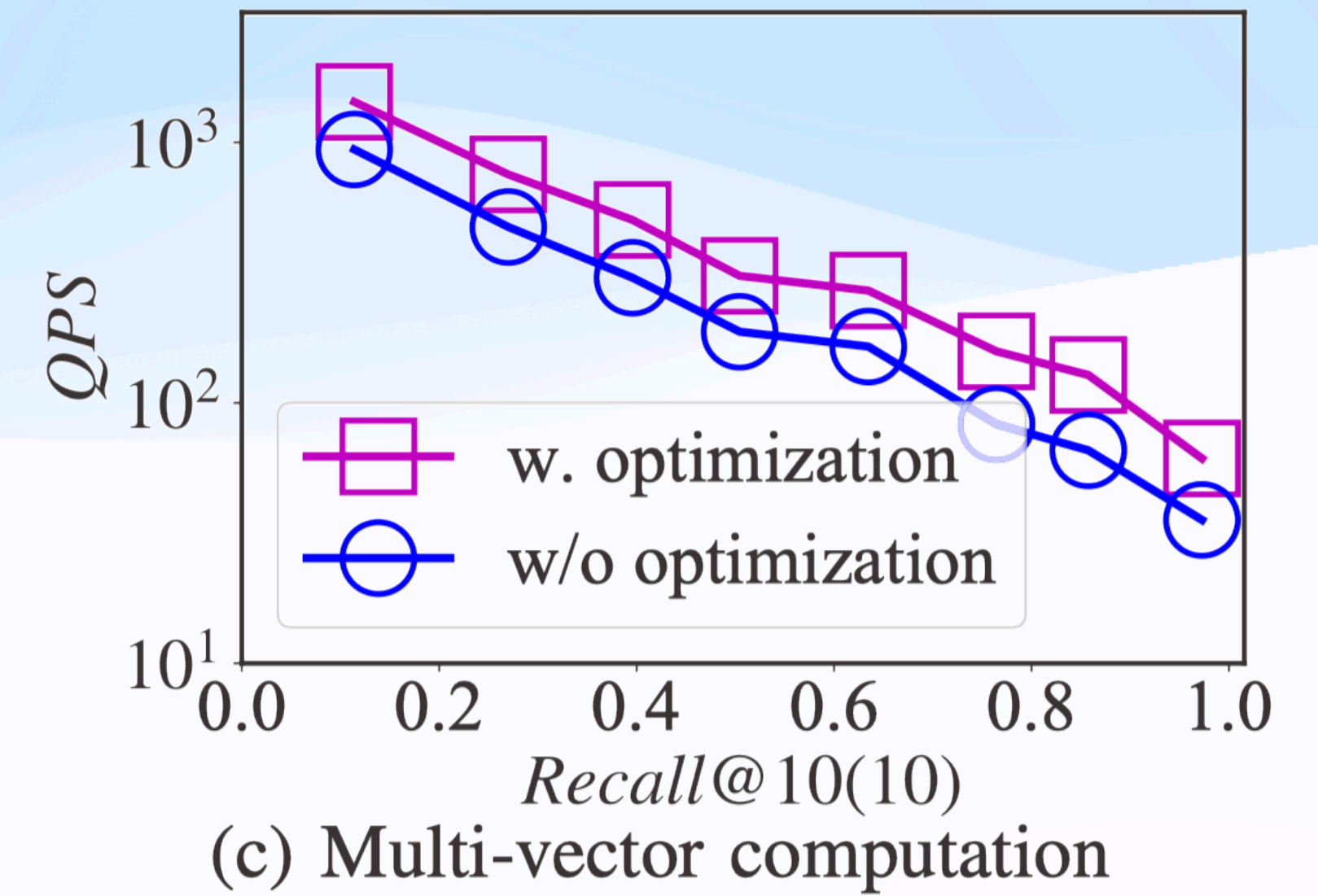
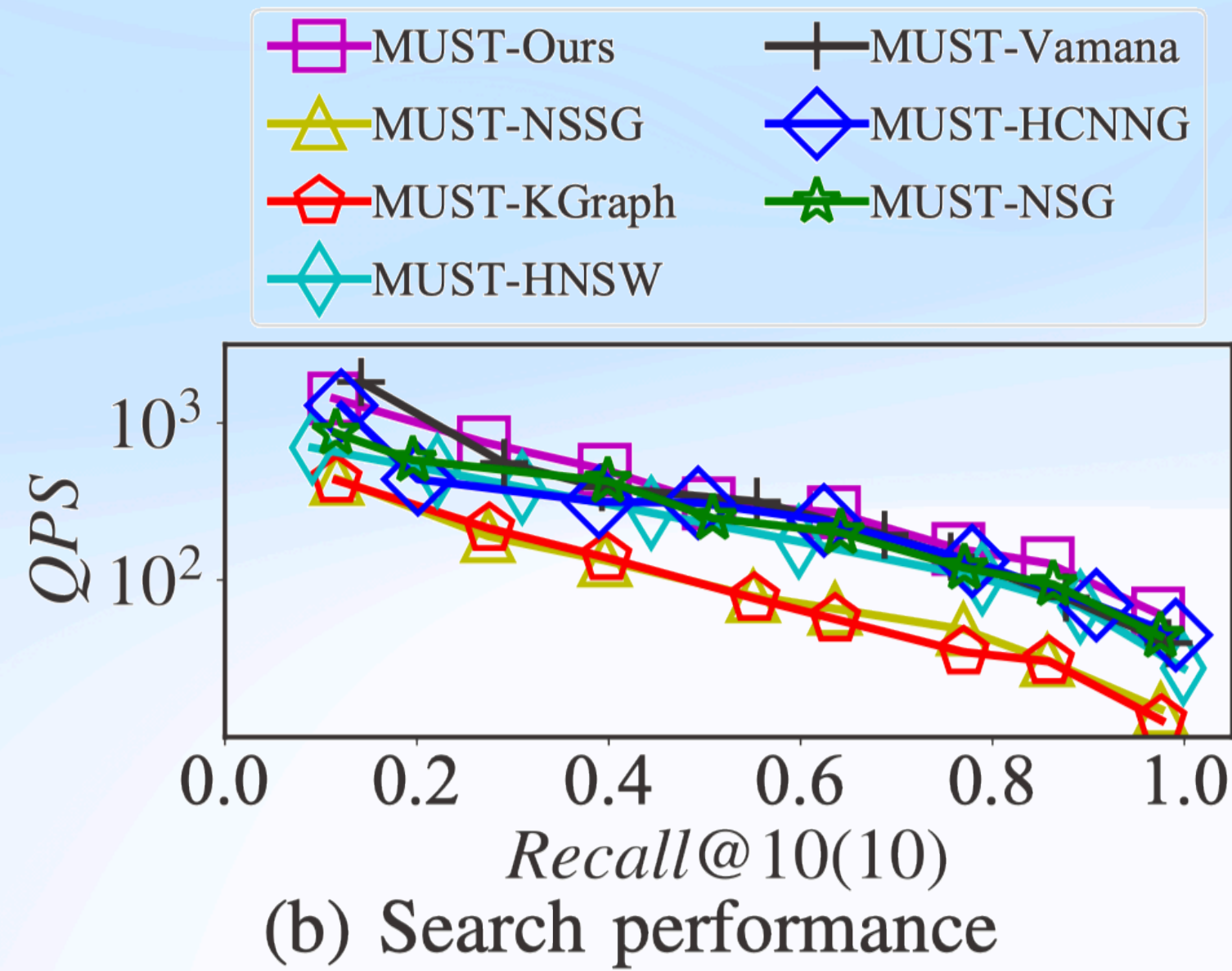
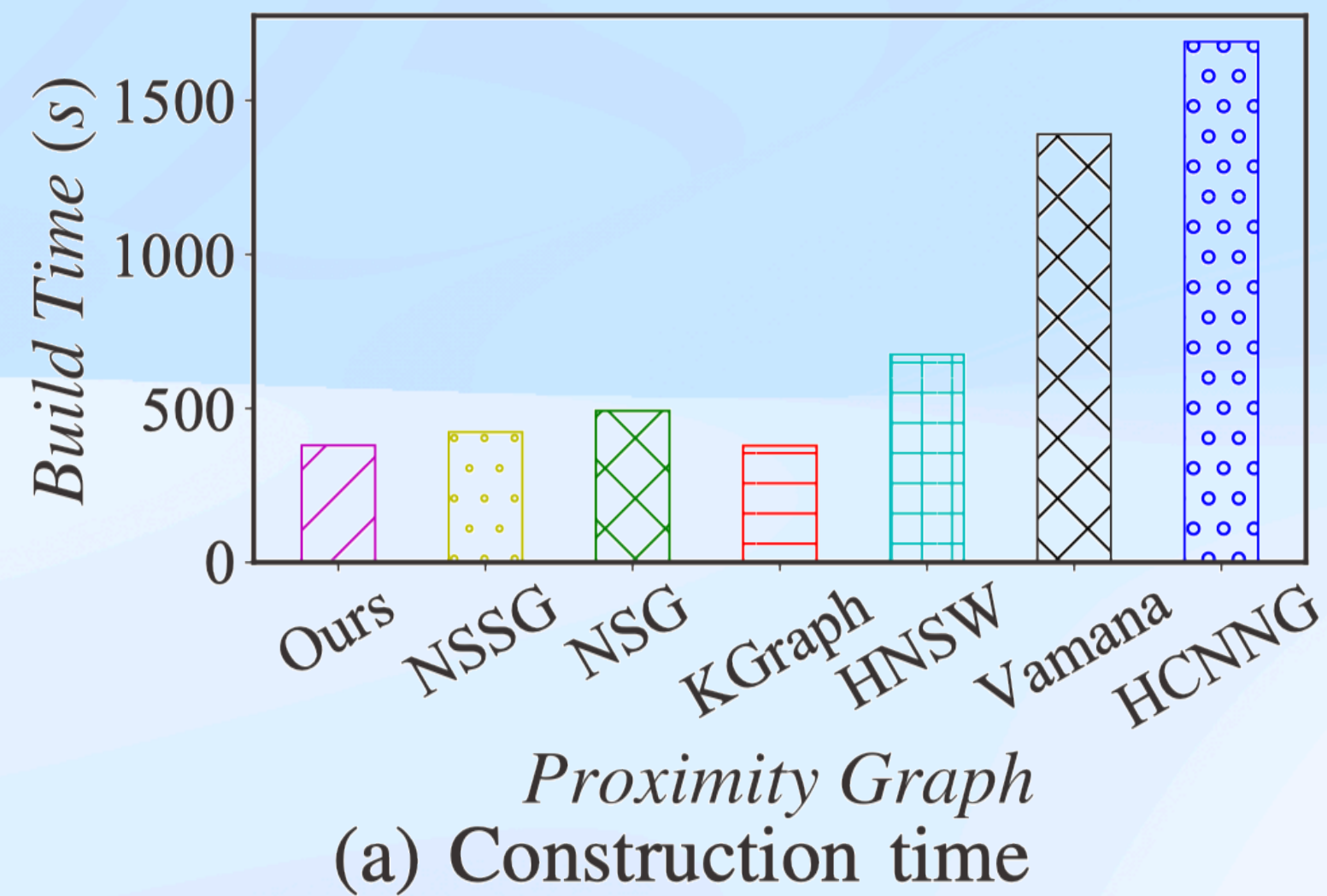
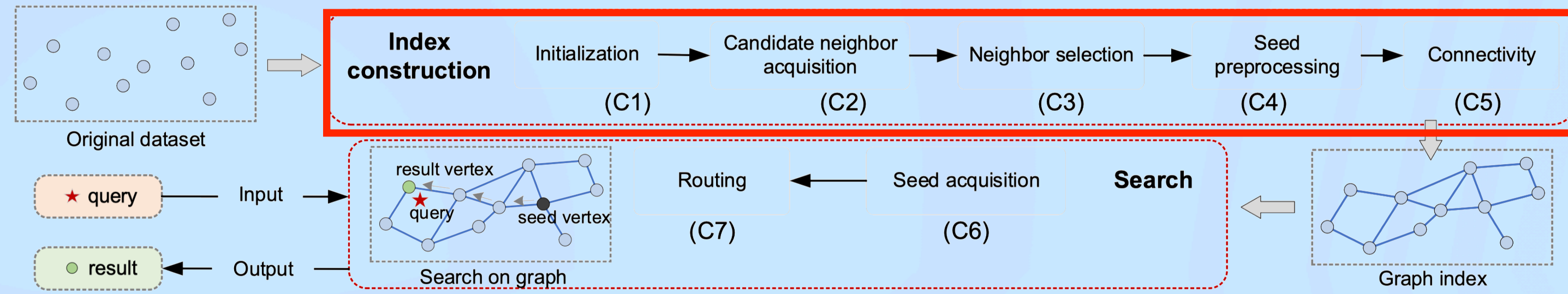
- MUST framework enable user to manually adjust weight of each modality.

Weights	ω_0^2	0.5	0.6	0.7	0.8	0.9
	ω_1^2	0.5	0.4	0.3	0.2	0.1
$IP(\phi_0(q^0), \phi_0(r^0))$		0.6915	0.7009	0.7440	0.8286	0.9301
$IP(\phi_1(q^1), \phi_1(r^1))$		0.9999	0.9960	0.9748	0.9242	0.8525

- However, user-defined weight might be harmful to recall rate.
- **MUST framework DO NOT care about your TARGET MODALITY.**

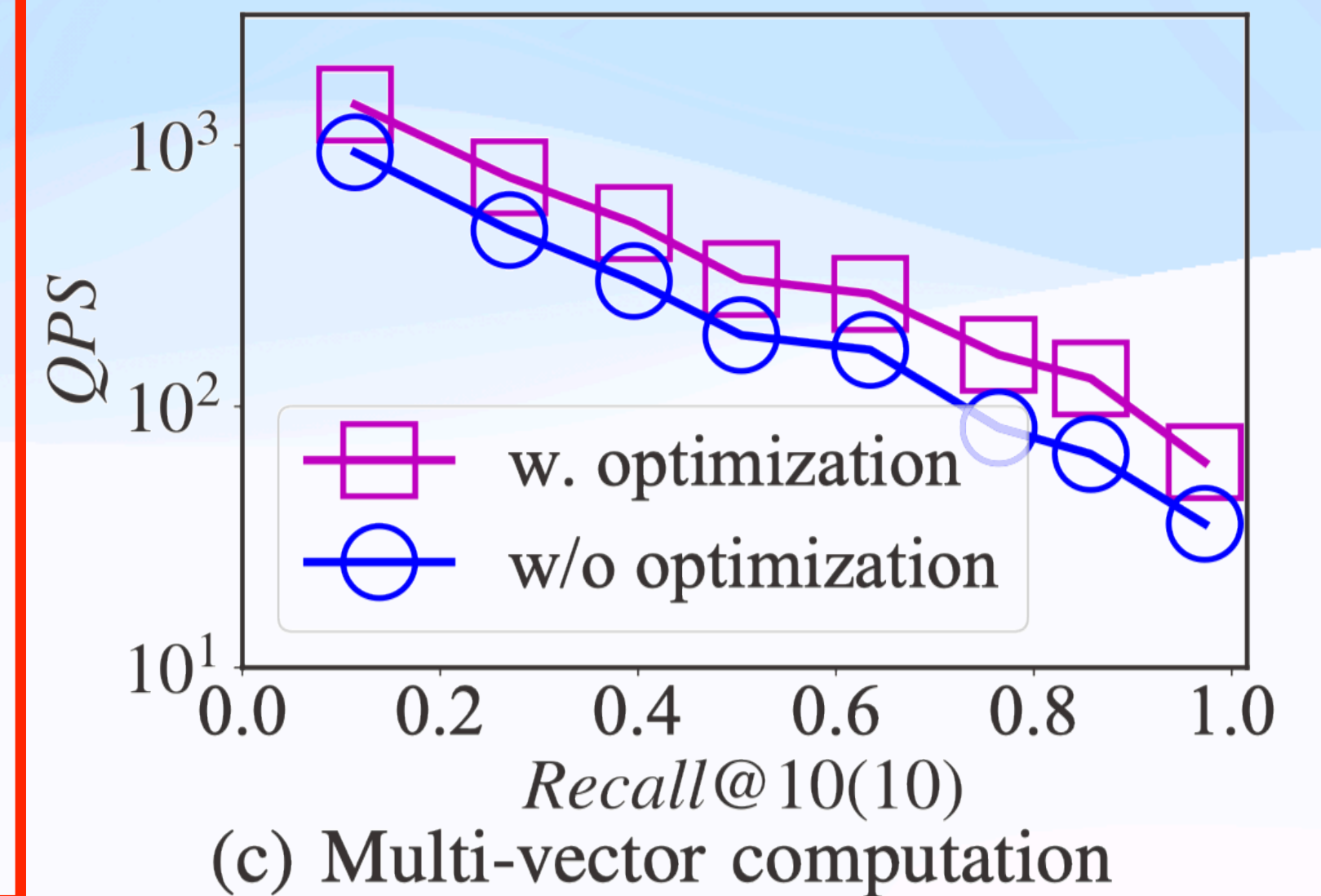
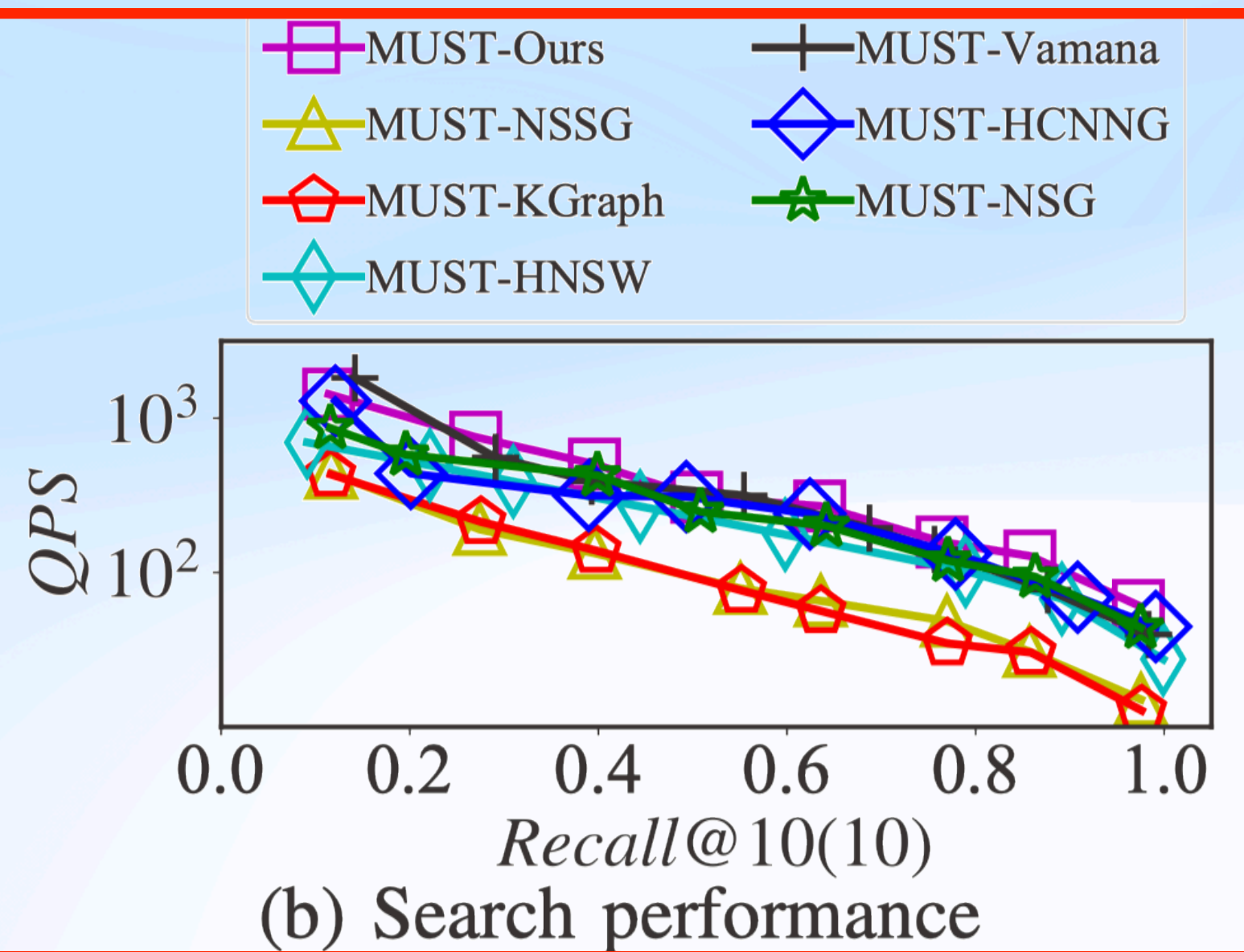
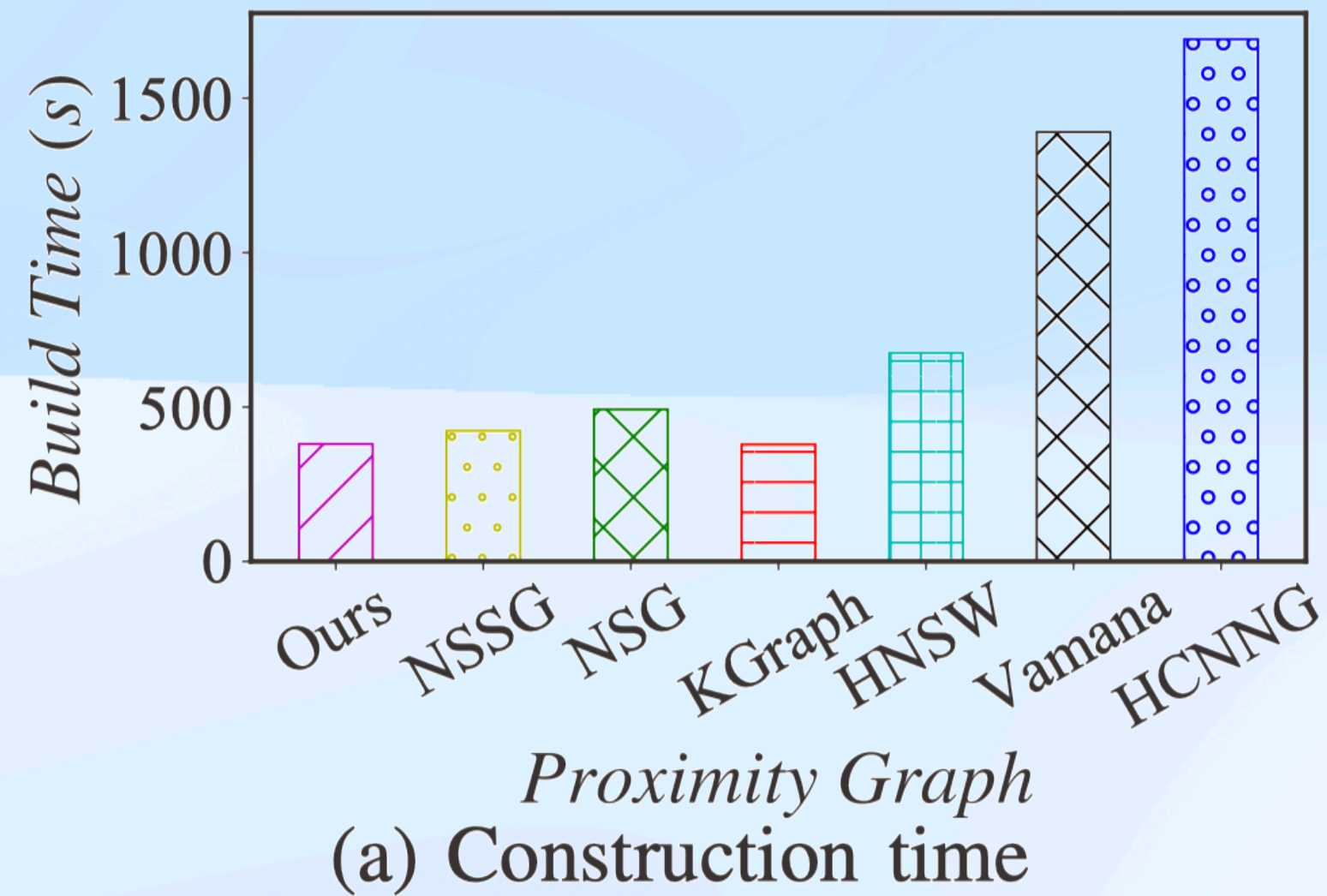
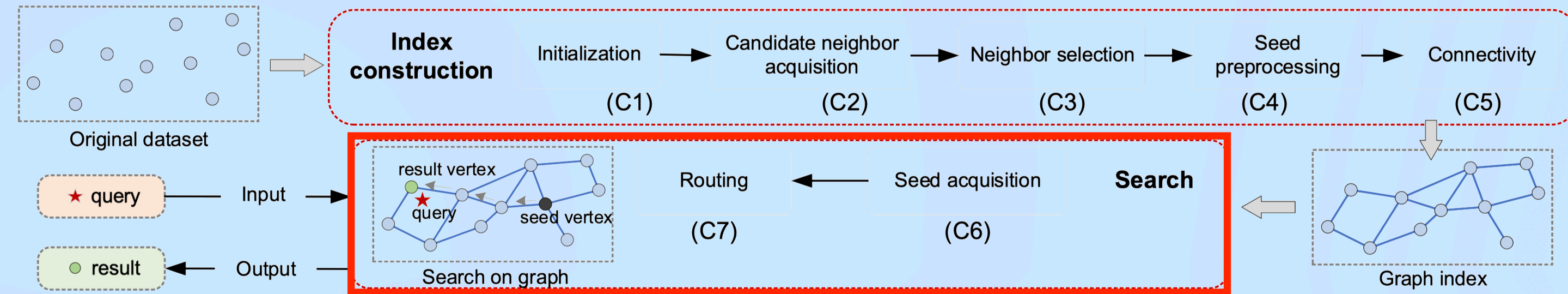
Evaluation

ANNS Algorithms



Evaluation

ANNS Algorithms



Evaluation

ANNS Algorithms

